

Power Demand Forecasting and Cost-Efficient Resource Allocation for Bangladesh's National Grid Using Machine Learning and Allied Techniques

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in partial fulfillment of the requirements for the degree of
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Declaration

It is hereby declared that

1. The thesis submitted is our own original work while completing degree at Brac University.
2. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

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Abstract

The rapid industrialization and population growth in Bangladesh have resulted in a significant increase in electricity demand, making energy efficiency a national priority. Given this expanding energy demand and limited generation resources, ensuring a reliable and cost-effective electricity supply has been a major challenge for the country's national grid. Traditional forecasting and planning methods often fail to capture the dynamic consumption patterns caused by population growth, weather conditions and industrial activity. To address this problem, this study develops an integrated framework that combines power demand forecasting with cost-efficient resource allocation for the national grid. Using the Power Grid Company of Bangladesh's (PGCB) hourly generation and demand records (2015-2025), together with weather, and macroeconomic features, several machine learning and deep learning models were trained and evaluated under both a chronological hold-out split and an eight-fold walk-forward validation scheme. A specific Random Forest / LightGBM configuration was found to forecast hourly demand most reliably, reaching a test MAPE of 4.16-4.47% across the evaluated settings. The resulting demand forecast was then used as the input to a ten-source economic load dispatch (ELD) model of the national grid (gas, liquid fuel, coal, hydro, solar, wind and four cross-border import links). For this, the capacity, availability and cost parameters were derived from the Bangladesh Power Development Board's (BPDB) official daily generation reports and Annual Report 2024-25. The dispatch problem was solved as a strictly convex quadratic program with a certified Karush-Kuhn-Tucker (KKT) optimum, and the same problem was independently re-solved with a Lagrangian-multiplier-based classical formulation, Genetic Algorithms (GA), Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), and an augmented-Lagrangian neural network (LANN), so that the optimisation paradigms could be compared on identical problems rather than in isolation. Re-allocating generation and the BPDB-reported fuel-availability constraint yields a model-estimated cost reduction of approximately 21% relative to the historical dispatch over the January-June 2025 study window (11-35% depending on gas availability). The result was cross-checked against the IEEE 30-bus benchmark system, the PyPSA-BD Bangladesh grid model and NREL's RTS-GMLC dataset. This figure is strictly a model-estimated reduction under the stated cost and availability assumptions, not a guaranteed real-world saving. The proposed framework is intended to benefit policymakers and power system operators by illustrating a transparent, reproducible and Bangladesh-specific method for cost-aware dispatch planning, thus contributing to more sustainable and data-driven power management in Bangladesh.

Keywords: Power demand forecasting, Machine learning, Resource optimization, Economic dispatch, Lagrangian multiplier, Genetic algorithm, Swarm intelligence, Ant colony optimization, Bangladesh national grid

Dedication (Optional)

A dedication is the expression of friendly connection or thanks by the author towards another person. It can occupy one or multiple lines depending on its importance. You can remove this page if you want.

Acknowledgement

Table of Contents

Declaration	i
Approval	ii
Ethics Statement	iv
Abstract	v
Dedication	vi
Acknowledgment	vii
Table of Contents	viii
List of Figures	xi
List of Tables	xiii
Nomenclature	xiii
1 Introduction	1
1.1 Background	1
1.2 Rationale of the Study or Motivation	2
1.3 Problem Statement	2
1.4 Objective	2
1.5 Methodology in Brief	3
1.5.1 Data Collection and Understanding	3
1.5.2 Feature Engineering	3
1.5.3 Data Preprocessing Pipeline	4
1.5.4 ML Models and Evaluation Metrics	4
1.5.5 Optimized Resource Allocation	4
1.6 Scopes and Challenges	5
2 Literature Review	7
2.1 Preliminaries	7
2.2 Review of Existing Research	9
2.3 Summary of Key Findings	13

3	Requirements, Impacts and Constraints	19
3.1	Final Specifications and Requirements	19
3.1.1	Functional Requirements	19
3.1.2	Data Requirements	20
3.1.3	Software and Hardware Requirements	20
3.1.4	Operational Scope	20
3.2	Societal Impact	21
3.3	Environmental Impact	21
3.4	Ethical Issues	21
4	Proposed Methodology	22
4.1	Methodology Overview	22
4.2	Forecasting Dataset Collection and Description	24
4.3	Forecasting Dataset Preprocessing	27
4.3.1	Data Cleaning	27
4.3.2	Data Transformation	28
4.3.3	Data Integration	29
4.3.4	Data Reduction	30
4.3.5	Summary of Preprocessed Data	31
4.4	Forecasting Models and Experimental Design	33
4.4.1	Models Evaluated	33
4.4.2	Validation Strategy	34
4.4.3	Development Environment	35
4.4.4	Preprocessing and Feature Engineering Pipeline	36
4.4.5	Evaluation Metrics	37
4.4.6	Experimental Strategy 1: Chronological Hold-Out Split	38
4.4.7	Hyperparameter Optimisation Framework	38
4.4.8	Experimental Strategy 2: Walk-Forward Temporal Validation	39
4.5	Optimisation Dataset Collection and Description	39
4.6	Economic Load Dispatch Problem Formulation	40
4.7	Optimisation Methods	40
4.8	Optimisation Validation and Benchmarking Methodology	41
5	Result Analysis	42
5.1	Performance Evaluation	42
5.1.1	Forecasting Evaluation Metrics	42
5.1.2	Optimisation Evaluation Metrics	42
5.2	Analysis of Design Solutions	43
5.2.1	Forecasting Results	43
5.2.2	Optimisation / Economic Dispatch Results	49
5.2.3	LANN Training Behaviour and Convergence	53
5.3	Final Design Adjustments	54
5.4	Statistical Analysis	55
5.5	Comparisons and Relationships	56
5.5.1	Benchmarking against independent systems	56
5.5.2	Relationship between forecast accuracy and dispatch cost	58
5.6	Discussions	58

6	Conclusion	60
6.1	Summary of Findings	60
6.2	Contributions to the Field	61
6.3	Limitations	62
6.4	Recommendations for Future Work	62
	Bibliography	64

List of Figures

1.1	Proposed methodology for power demand forecasting and cost-efficient resource allocation for Bangladesh's national grid	5
4.1	Workflow Diagram of the Study	24
4.2	Correlation of Lag and Rolling Features with Target Demand	29
4.3	Spearman rank correlation between each engineered feature and the target demand_mw. Lag and rolling features dominate the top ranks, confirming the strong autoregressive structure of the demand series.	33
4.4	End-to-end preprocessing and modeling pipeline.	37
4.5	Optimization Pipeline	40
5.1	Actual versus predicted electricity demand (MW) produced by LightGBM on the 2024–2025 test period. The weekly zoom inset illustrates the model’s ability to reproduce diurnal peak–trough cycles and weekend demand suppression.	44
5.2	Test MAPE for each walk-forward fold (Folds 2–8) across all seven traditional ML models. The horizontal axis represents the test year (2019–2025). Gray shaded band indicates the range (min–max) of the four tree-based ensemble models (Random Forest, Gradient Boosting, XGBoost, LightGBM) per fold.	47
5.3	Comparison of test MAPE on the 2025 test period between the average of Folds 2–8 (2-year training windows) and the Custom Fold (4-year training window) for Random Forest and LightGBM. The substantial reduction in MAPE (from $\approx 4.2\%$ to $\approx 2.7\%$) demonstrates the benefit of incorporating a longer, more recent four-year training window.	48
5.4	<i>Algorithm comparison over the 24-hour stratified sample — total cost distribution (GA/PSO/ACO, 30 runs), mean optimality gap vs. the QP reference, runtime, and mean power-balance error</i>	49
5.5	<i>Model-estimated cost reduction over the study window — 7-day rolling mean over time (left) and hourly distribution with mean 21.28% (right), under the base-case (B2) availability assumption</i>	51
5.6	<i>Hourly dispatch by source, actual versus optimised</i>	52
5.7	<i>Generation mix by source, actual vs. optimised energy share</i>	52
5.8	<i>Weekly and monthly actual vs. optimised dispatch cost trend across the study window</i>	53
5.9	<i>GA/PSO/ACO convergence over the optimisation budget</i>	54
5.10	<i>LANN training history — cost, mean constraint violation, penalty weight (ρ) and mean dual variable (λ) versus training epoch</i>	54

5.11	<i>External benchmarking - PyPSA-BD vs. this study's domestic capacity share (left) and IEEE 30-bus optimality gaps with genuine cost coefficients (right)</i>	56
5.12	<i>Sensitivity of the estimated cost reduction to the quadratic cost coefficient (κ), import cost basis and coal scenario</i>	57
5.13	<i>Availability scenarios and BPDB-reported daily fuel cost series used to derive the model's cost and capacity parameters</i>	57
5.14	<i>Forecast-error robustness (regret vs. forecast bias) and the LANN's learned dual variable tracking the true system marginal cost</i>	58

List of Tables

2.1	Key studies from the reviewed literature	14
3.1	Software library requirements	20
4.1	PGCB Dataset Sample: Core Columns	25
4.2	PGCB Dataset Sample: Domestic Generation (MW)	25
4.3	PGCB Dataset Sample: Imports (MW)	25
4.4	Original PGCB dataset fields, their descriptions, and their treatment in the preprocessing pipeline.	26
4.5	Trigonometric cyclical encoding of temporal features applied to all hourly observations.	28
4.6	Final integrated feature matrix composition after all collection, cleaning, transformation, and integration steps.	30
4.7	Chronological train-validation-test partition used in the base pipeline. . .	31
4.8	Summary of Final Preprocessed Dataset	31
4.9	Complete Feature List — Category, Name, and Description	31
4.10	Walk-Forward Validation Setup Across Folds	35
4.11	Software Libraries and Frameworks	35
4.12	Statistical Evaluation Metrics	38
5.1	Experimental Strategy 1 — Full Model Comparison (Chronological Split)	43
5.2	LightGBM Optuna Hyperparameter Search Space and Optimal Configuration	44
5.3	LightGBM Baseline vs. Optuna-Tuned Performance	45
5.4	Walk-Forward Validation — Per-Fold Test MAPE (%) for All Models . .	45
5.5	Walk-Forward Summary — Folds 2–8 Average Performance	46
5.6	Custom Fold Results — Training 2020–2023, Validation 2024, Test 2025	47
5.7	Main dispatch method comparison (24-hour stratified sample, 30 runs for GA/PSO/ACO)	49
5.8	Model-estimated cost reduction under different fuel-availability assumptions	50
5.9	Generation mix, actual vs. optimised (energy share, %)	51
5.10	Monthly cost reduction trend	53
5.11	Design iterations and their effect on the result	54
5.12	Friedman and Holm-corrected Wilcoxon test results (30 runs, 24-hour sample)	56
5.13	Forecast-error robustness (regret relative to perfect-foresight dispatch) . .	58

Chapter 1

Introduction

1.1 Background

Electricity has become an integral part of modern society. Manufacturing, healthcare, transportation, communication, digital services, and almost all the sectors of a country's economy are dependent on a reliable supply of electricity [1]. Furthermore, as countries go through their development arc, regardless of their stage, everyone will experience a significant increase in their demand for electricity [2]. For sustaining economic growth and improving the quality of life, it becomes a critical factor to generate the required amount of electricity, and that too in the most efficient fashion possible [3].

Bangladesh is no stranger to this phenomenon. It is among the fastest-growing economies as well as one of the fastest-growing in terms of population. Thus, the electricity demand has seen a drastic hike, and it is not slowing down [4]. To account for this demand, the Power Grid Company of Bangladesh (PGCB) relies on a mix of different energy sources. These sources include gas, liquid fuel, coal, hydropower, and renewable sources like solar and wind [5]. The increased demands can not be satisfied by local efforts. As a result, electricity has to be imported from neighboring countries [6]. Even though this satisfies the need for electricity, it makes the overall system significantly more complex.

One of the biggest challenges faced by PGCB is to forecast the demand accurately and generate the exact amount of electricity to satisfy the demand. If the forecast is a number below the actual demand, to compensate, load shedding will occur. If the forecast number exceeds the demand, the extra electricity generated will be wasted [7]. None of these situations is ideal, as the cost of the extra electricity will be counted as a loss. If the generated electricity is less than the demand, there will be no cost of wasted generation, but there will be discomfort for the citizens. It will hurt the economy as its related activities rely heavily on electricity [8]. Furthermore, the power sector will incur financial loss as the electricity that was not generated was caused by forecasting inaccuracies rather than capacity limitations [9]. Thus, it is very crucial that the forecasting is as accurate as possible for both the quality of life and the betterment of the economy.

With the recent developments in smart grid technology, the grids can now take in great amounts of both internal and external data, process them, and forecast more accurately [10]. Depending on situations, the grid system can take different decisions or even choose a different machine learning model to best address the situation. The PGCB publishes de-

tailed hourly electricity generation and demand data, along with the amount of electricity sourced from various sources [5]. This data, along with other external data and machine learning, could open up a plethora of possibilities in the context of Bangladesh's power grid, and that is what we are planning to explore.

1.2 Rationale of the Study or Motivation

Bangladesh's power sector is facing increasing amounts of financial and operational pressure due to the heavy reliance on imported items to meet demand. Imports are not limited to imports of electricity from neighboring countries, but also extend to the raw materials needed to generate electricity locally [8]. If the demand forecasting is inaccurate, it can lead to higher generation costs and, in turn, increased pressure on the economy. That is why it is absolutely necessary to make the forecast accurate, which in turn improves overall operational efficiency.

As mentioned before, the different sources of energy, their capacity, other limitations, and variables make it very complex and challenging to figure out what sources to use, and how much to use to generate the required amount of electricity at the least possible cost. A system that could optimize this mix in a cost-optimal manner would mean faster and more accurate decision-making, depending on the demands [7]. The grid will know what would be the best possible mix to minimize cost, as well as a long-term demand picture to budget and plan what and how much of it needs to be imported. The availability of the PGCB hourly data and of detailed operational and cost data provides an opportunity to create such a system.

1.3 Problem Statement

The power grid of Bangladesh must account for supply and demand under fluctuating power consumption, variable renewable energy, and rely on both domestic and imported energy and raw materials. Not only this, but it must also handle the complexity of the ever-changing capacities, prices, availability, and other factors of all these sources of electricity. This creates a highly complex operational environment [5]. It requires having proper systems in place to come up with highly accurate decisions to operate efficiently under these constraints. Although Bangladesh has adopted AI-based solutions, they focus primarily on operational stability rather than cost-optimal generation planning [7]. However, a system derived from the grid's own data to accurately forecast electricity demands and suggest the mix to satisfy that demand at the minimum cost is yet to be made. This thesis addresses this gap by developing a machine-learning-based forecasting model using PGCB's hourly generation data, along with other engineered features, to further refine forecasting accuracy and estimate electricity demand. In addition, the system will infer the most cost-effective mix to meet the projected demand.

1.4 Objective

The primary objective of our research is to develop a framework to predict the electricity demand, and cost-efficient resource allocation for electricity generation to meet that

demand for the national power grid of Bangladesh. This would help the respective authorities to maintain a sustainable, and efficient power system planning in Bangladesh.

This study aims to:

- Analyze electricity generation pattern using the PGCB dataset.
- Improve demand forecasting by including seasonal, weather-related features through feature engineering.
- Implement and compare multiple ML models to find the best performing model to predict the demand most accurately.
- Introduce a cost-effective resource allocation technique to find the optimal generation mix to meet the predicted demand.

1.5 Methodology in Brief

The methodological workflow will be conducted in two parts for this research. First, the methodological process will be for power demand forecasting. Then, we will continue with the resource allocation using optimization techniques in a way that minimizes the production cost.

1.5.1 Data Collection and Understanding

The PGCB Hourly Generation Dataset (Bangladesh) [11] will be used for this research. This dataset is collected from the UCI machine learning repository. It has 14 features which are real, categorical, and integer values. This dataset provides hourly records of electricity generation, demand, loadshedding, and breakdown of generation mix which supports the regression task of forecasting demand.

1.5.2 Feature Engineering

To improve the accuracy of forecasting, several feature engineering techniques will be applied to the dataset. We can integrate day type features to indicate working days, weekends, and public holidays. We can also add seasonal information (summer, monsoon, winter). These data can be extracted from official government calendars. Then we can incorporate weather based features such as temperature, humidity, rainfall data. These data can be collected from publicly available sources such as the Bangladesh Meteorological Department (BMD) or global reanalysis datasets. Furthermore, we can add lag features, and rolling statistics. Lag features means using previous demand data as input. For example, the demand of electricity before 168 hours (lag-168) gives the demand for the same hour of last week. And rolling statistics gives valuable information about demand over a period of time. For example, rolling mean gives an average demand over a specified period. And rolling standard deviation helps us understand the variability of demand over time. These features will help us observe repeating patterns, and recent trends of demand.

1.5.3 Data Preprocessing Pipeline

A structured preprocessing pipeline will be maintained to prevent overfitting, and data leakage. First, we can handle missing values by using techniques like forward fill, or imputation. Then we can split the dataset for training and testing. Here the training set will contain the data of earlier time periods, and the testing set will contain the data of later time periods for determining accuracy of the predicted demand. After that we can perform normalization, and scaling on the training data and learning parameters will then be applied to the test data to prevent data leakage. Finally, we can select features using feature importance techniques, and remove highly correlated features to reduce multicollinearity.

1.5.4 ML Models and Evaluation Metrics

Based on recent studies, we will implement several ML models to forecast power demand. We can use traditional machine learning models like Linear Regression, and tree based ensemble methods like XGBoost. We can also apply deep learning based time-series models such as LSTM, GRU, or TiDE. The models will be trained on the processed dataset, and the results of the models will be compared to identify the best performing model. We will use the widely used error metrics to evaluate the forecasting performance such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), Coefficient of Determination (R^2).

1.5.5 Optimized Resource Allocation

After the prediction of electricity demand, optimization strategies will be introduced to find the optimal generation mix to meet that demand. The target of optimization will be to minimize the production cost while respecting the capacity of the resources. To achieve this goal, we can use Lagrangian multiplier based economic dispatch formulation. Several alternative optimization techniques will also be explored to handle complex cost functions and system constraints such as Genetic Algorithms (GA), Ant Colony Optimization (ACO), Particle Swarm Optimization (PSO), and other swarm intelligence methods. The comparative performance of these methods will help us identify the most appropriate optimization strategy for Bangladesh's power generation mix.

The outcome of this methodology, also shown in Fig. 1.1 will be an integrated framework which forecasts the electricity demand accurately, and provides an optimal generation mix.

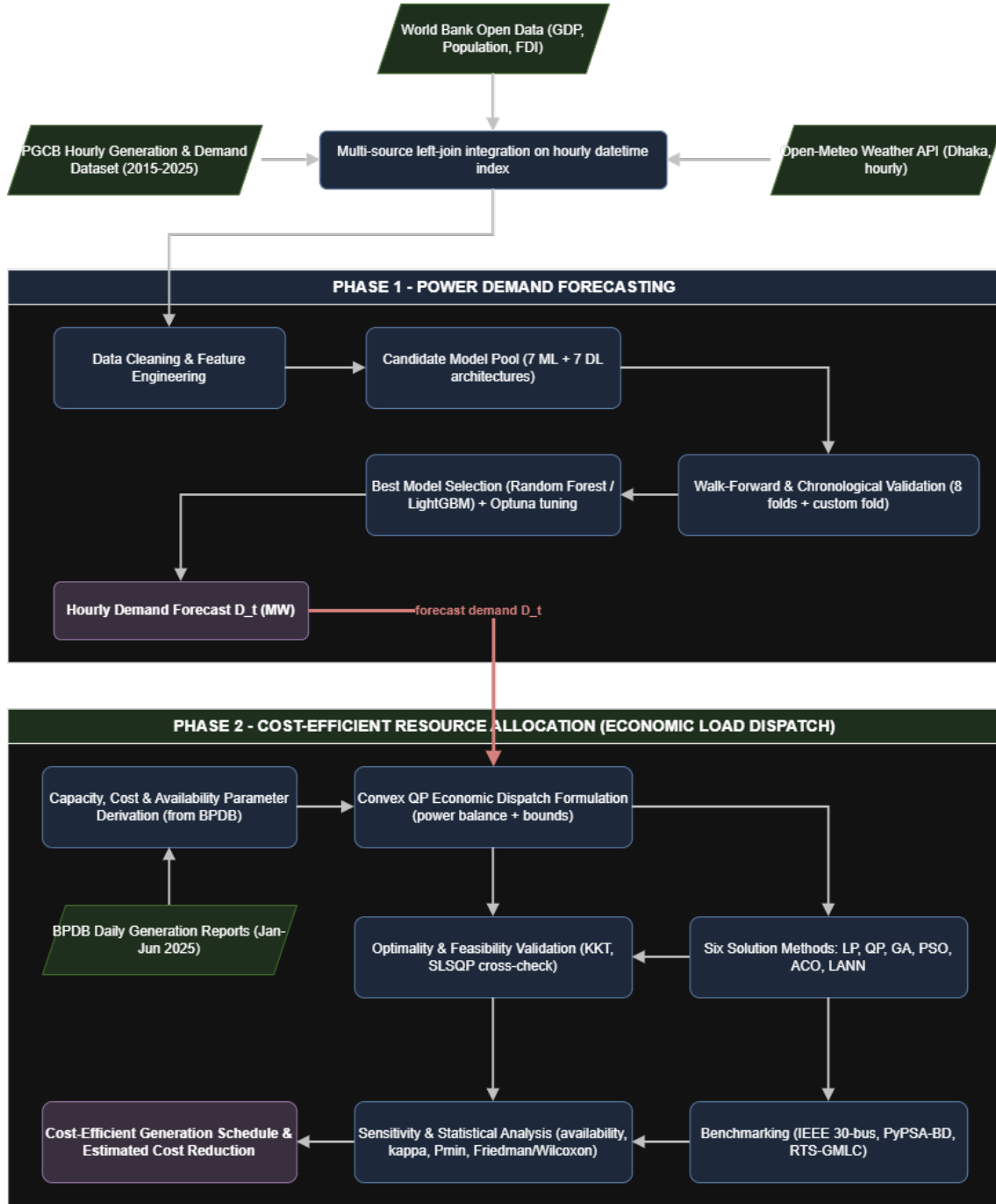


Figure 1.1: Proposed methodology for power demand forecasting and cost-efficient resource allocation for Bangladesh's national grid

1.6 Scopes and Challenges

Focusing on the major aim of the national power grid of Bangladesh which is to reduce the operational inefficiency, this study uses the advanced machine learning algorithms for predicting the load along with mathematical optimization for the efficiency. There exists many systems that still struggle with the load shedding and energy mixture problems. The goal is to develop a pipeline so that it can accurately forecast the potential demand of the loads and applies different optimization frameworks such as Lagrangian Multipliers

to ensure the most cost effective and stable power distribution using an efficient energy mix from renewable sources and cross border imports.

Scope of the Study: Though there are many scopes of solving the national power problem, this research sets boundaries to reflect the technical and environmental realities of the power sector in Bangladesh:

- **Administrative Focus:** This study covers the thermal, hydro, renewable sources along with imports of energy from different international sources which is only limited to the national power grid managed by PGCB.
- **Temporal and Data Range:** The research utilizes a decade of historical records (2015–2025), capturing the significant transitions in Bangladesh's energy landscape, including the recent integration of large-scale coal and imported power.
- **Methodological Framework:** The scope includes the implementation of specific architectures, primarily Bidirectional GRU (BiGRU) and TiDE for forecasting, followed by the application of Lagrangian Multiplication methods and metaheuristic techniques for economic load dispatch.
- **Environmental Integration:** There are extreme sensitivity of in the context of power grid of Bangladesh due to temperature and other weather spikes. This study uses high-resolution meteorological data to reduce the sudden changes.

Challenges Encountered: This study have different challenges to overcome for the essential national grid stability.

- **Hidden Demand Gap:** One of the major problem in the historical record is that is often predicts the demand from served demand data rather than a true demand required. The actual demand usually covered up by the continuous load shedding which causes difficulty in model training leading to a gap in the energy demands.
- **Extreme Weather Sensitivity:** The non-linear relationship between temperature extremes and power load in Bangladesh is difficult to model. Sudden heatwaves can cause demand to spike beyond the training parameters of standard algorithms.
- **Optimization Complexity:** Using Lagrangian Multipliers to balance cost and stability involves managing a high number of constraints, such as generator ramp rates and prohibited operating zones, which can lead to slow convergence in real-time scenarios.
- **Non-Stationarity of Infrastructure:** With the fast transformation of power sources (i.e. Adani, Nepal) the challenge of coping with the data is also increasing for the models. Because historical data patterns just few years ago is no longer usable with this rapid change.
- **The "Black Box" Problem:** Though deep learning gives a high accuracy in terms of specific grid power transmit, the underlined mechanism is not clear in because of the deep hidden layers. This creates a trust and adoption barrier in the real world adoption of specific power generation dispatch.
- **Fuel Data imbalance:** Generation capacity can be impacted greatly by the availability of primary sources (i.e natural gas) which cannot be determined only by historical data

Chapter 2

Literature Review

2.1 Preliminaries

Researchers have adopted different advanced methods of economic efficiency and power stability in the complex and dynamic landscape of national power grid and energy management of Bangladesh. By upgrading from traditional statistical methods to advanced machine learning and metaheuristic optimization frameworks, it allowed the Power Grid Company of Bangladesh (PGCB) to address challenges of efficient power management [11]. These methods are used for predicting load up-downs, optimizing generation mixture, minimizing costs and integrating renewable energy sources.

Long Short-Term Memory (LSTM) networks serve as the foundational architecture for electricity demand forecasting in Bangladesh[12]. This method is useful for its effectiveness of capturing long-term dependencies in the time-series data which is a must for hourly and daily load predictions. For further enhancement, Bidirectional LSTMs (Bi-LSTMs) along with Gated Recurrent Units (GRU) are used for processing data from both directions at the same time helping the models to understand the context deeper for hourly demand spikes. Also, hybrid models like CNN-BiLSTM are implemented for complex urban environments like Dhaka or Sylhet [13]. Here CNN extracts temporal features that is relevant for a particular timeframe and Bi-LSTM stacked layers manage long-term dependencies.

Advanced architectures like the Time-series Dense Encoder (TiDE) have emerged to address the noise and randomness inherent in rapidly developing regions. TiDE utilizes dense multi-layer perceptron (MLP) structures to achieve higher precision across various forecast horizons which outperforms traditional linear regression and ensemble models [14]. In scenarios where tabular data is primary, such as analyzing the impact of temperature, summer/winter transitions, and holiday tagging, gradient boosting methods like LightGBM are utilized [15]. LightGBM is particularly valued for its ability to handle non-linear seasonal relationships more effectively than statistical baselines like SARIMA [15].

The Lagrangian Multiplier approach is used to handle the more complicated mathematical optimization. These strategies allow the model to break down challenging grid issues, such the overall generation vs overall demand into smaller and more manageable challenges for each power plant [16]. The Lagrangian method's basic idea is to turn a constrained problem to an unconstrained one. To minimize an objective function $f(x,y)$

subject to a constraint $g(x, y) = c$, a new **Lagrangian** ($\mathcal{L}$) function is created.

$$\mathcal{L}(x, y, \lambda) = f(x, y) + \lambda \cdot [c - g(x, y)]$$

Here, λ (lambda) is the **Lagrangian Multiplier**. The stationary points of this new function (where its partial derivatives are zero) provide the solution to the original constrained problem [16]. This strategy works best for large-scale systems like PGCB where the main goal is to minimize the fuel cost but considering safety limits. The optimization of the power grid is primarily addressed through the most famous application of the Lagrangian Multiplier method, the Economic Load Dispatch (ELD) problem [16]. The Lagrangian for this system is written as:

$$\mathcal{L} = \sum C_i(P_i) + \lambda(P_{load} + P_{losses} - \sum P_i)$$

where, $C_i(P_i)$ is the cost function for generator i . To get the Increment Cost Criterion, the derivative of $\mathcal{L}$ is performed with respect to each power output P_i and set it to zero:

$$\frac{dC_1}{dP_1} = \frac{dC_2}{dP_2} = \dots = \lambda$$

The fundamental principle from the derivative shows that all the generators that is removable should operate at the exact incremental cost (λ) for getting the minimum power grid operating cost. These techniques of metaheuristics consider the constraints like ramp rate limits and Prohibited Operating Zones (POZ) [16].

Apart from the first-order stationarity argument, the ELD problem can be posed directly as a constrained quadratic program when each generator's cost function is modelled as convex quadratic, $C_i(P_i) = a_i + b_i P_i + c_i P_i^2$. In this formulation, the Karush-Kuhn-Tucker (KKT) conditions are both necessary, and as the problem is strictly convex, sufficient for global optimality [17]. This provides a certifiable optimum against which any metaheuristic or learned method can be benchmarked. The present study uses this property directly where the KKT stationarity conditions are solved by a bisection-and-Newton search over the shared marginal cost (the dual variable, λ), and the resulting dispatch is independently cross-checked with a general-purpose sequential quadratic programming solver (SLSQP) [18].

A further collection of methods, known as learning-to-optimize or amortised optimisation, trains a neural network to output a feasible, near-optimal dispatch directly from the problem instance, replacing an iterative solve with a single forward pass once finished training. Such networks are commonly trained with an augmented-Lagrangian (method-of-multipliers) loss, in which a learned or updated dual variable enforces the equality constraint while a quadratic penalty term stabilises training [19]. This study implements such a Lagrangian neural network (LANN) as a sixth dispatch method, alongside the Lagrangian-based classical formulation, the QP reference, GA, PSO and ACO, so that a learned optimiser can be compared against exact and metaheuristic baselines on the same economic dispatch problem.

Also, there are other optimization algorithms used widely which includes Particle Swarm Optimization (PSO), Harmony Search (HS), Ant Colony Optimization (ACO) etc. These optimization falls under the umbrella of Swarm Intelligence (SI). Unlike Lagrangian Multiplier method it doesn't rely on calculus and gradients, rather it is nature-inspired and it relies on simple agents interacting locally with one another and their environment. Since

the power grid problems are often non-linear, non-convex and noisy, SI performs very well. The PSO is used for cost and loss minimization. In this technique, the solution is obtained from observing particles flying through a multi-dimensional search space [20]. Each particle remembers its own best position found so far (personal best) along with the best position found from all the members of the entire swarm (global best). The velocity of the particles are updated by

$$v_i(t+1) = wv_i(t) + c_1r_1(p_{best,i} - x_i(t)) + c_2r_2(g_{best} - x_i(t))$$

and the position is updated by

$$x_i(t+1) = x_i(t) + v_i(t+1)$$

where w , c and r represent initial weight, acceleration coefficient and random number respectively. In the use of PSO in power systems, the search space is created by a mathematical axes of the generator outputs, bus voltages and transformer settings [20].

ACO, on the other hand, is inspired by the Ant behavior pattern for finding the shortest distance between their nest and a food source guided by chemical signals (e.g. Pheromone). If an ant k moves from node i to node j , the probability (P_{ij}^k) is calculated by

$$P_{ij}^k = \frac{[\tau_{ij}]^\alpha \cdot [\eta_{ij}]^\beta}{\sum [\tau_{il}]^\alpha \cdot [\eta_{il}]^\beta}$$

Here τ_{ij} represents pheromone intensity, η_{ij} is heuristic desirability and α, β are the control units. This method is widely used for solving combinatorial optimization problems and also in generator systems [21]. Instead of finding shortest path, it is used for decision making of the generator (ON/OFF), network configuration for minimal power loss, deciding where to place capacitor on the grid for voltage stability etc.

Again, Genetic Algorithm (GA) uses mechanism to replicate the natural selection and genetics to find the best possible solutions. This algorithm uses techniques like taking random initial population (candidate solution), defining fitness function, selecting best genes, crossover to get the best traits from the parent and mutation for diversity. In the power grid system, GA is used to reduce the massive exponential complexity from the long-term planning (e.g 20 years) problems, to control scheduling of the generators and for finding best configuration for optimal power flow. HS is often used to prioritize the speed of computation for real time dispatch. Also as an open source tool, Open Source Energy Modelling System (OSeMOSYS) is vastly used for its free, user-friendly and powerful structure of modeling long term energy planning by simulating different scenarios [22].

2.2 Review of Existing Research

The literature review explores recent studies on power demand forecasting and cost-efficient resource allocation. The studies that used Machine Learning, Deep Learning, and optimization approaches were emphasized.

To begin with, A 2025 study [23] applied ML models to predict power demand and fuel consumption for a 150 MW heavy fuel oil power plant in Bangladesh. This study used 7

years (2017 - 2023) of operational data including working hours, power generation, auxiliary consumption, atmospheric temperature, relative humidity, and atmospheric pressure. Data cleaning, and standardization were performed in the data preprocessing phase. The research implemented three ML models; K-Nearest Neighbors (KNN), Artificial Neural Networks (ANN), and Gradient-Boosted Regression Trees (GBRT) to predict the power demand and fuel usage. Coefficient of Determination (R^2), and Root Mean Square Error (RMSE) were used as evaluation metrics. And GBRT was identified as the best performing model with highest accuracy, R^2 , and lowest RMSE. However, this study used only one plant's data. Future work includes broadening the dataset with data from multiple plants, and linking demand prediction with generation dispatch, and cost optimization.

Another study [15] explored how environmental and seasonal variables can influence the electricity consumption in Dhaka city. It proposes a solution based on the STLTF (short-term load forecasting) framework. It uses daily electricity demand data from BPDB (Bangladesh Power Development Board) and combines it with additional temperature and seasonal data to demonstrate the role of these features in influencing electricity demand. A great contribution of the paper in question is feature engineering. It incorporates variables such as public holidays, religious festivals, and humidity to better capture their impacts on the demand. Here, the authors compared traditional time-series models (SARIMA) with modern machine learning techniques such as LSTM and LightGBM. According to their tests, LightGBM performs the best, specifically when there are non-linear demand spikes. Their results show a significant reduction in forecasting error from the original dataset. However, this research is limited to an urban context and a single city. In contrast, our thesis tries to scale this to a national level, while adding more features and techniques to fine-tune the results and supporting operation-level decision making rather than only analysis.

Moving forward, in another study [24], the researchers explored the use of AI to actively control and optimize energy generation in rural microgrids in Bangladesh. The authors focused on regions that have weak grid connectivity, and electricity is supplied, or demand is met through multiple sources such as solar systems, diesel generators, and grids. One key finding of this study is that their system significantly cut down on diesel usage, in turn cutting down on its cost and emissions. This positively impacted all the inhabitants of the area directly. However, this study is limited to a very small area in Bangladesh. Our thesis builds upon the same core idea as this, but the scale and complexity will be much higher.

Furthermore, addressing the volatility and non-linearity of very short-term load forecasting (VSTLF), Yang et al. [25] proposed a hybrid deep learning model that integrates Convolutional Neural Networks (CNN) with Bidirectional Long Short-Term Memory (BiLSTM) architectures. The model employs a CNN layer to extract spatiotemporal features from historical power load, temperature, and humidity data, followed by a BiLSTM network to model long-term and short-term temporal dependencies [25]. To ensure model robustness, the researchers utilized Min-Max scaling for normalization and Pearson correlation coefficients for attribute selection [25]. A standout feature of this research is the integration of Shapley Additive Explanations (SHAP), which provides a quantitative ranking of feature importance, thereby improving the interpretability of the deep learning "black box" [25]. The results indicate that the CNN-BiLSTM configuration achieves su-

perior performance compared to traditional models, yielding a Root Mean Squared Error (RMSE) of 49.98 and a Coefficient of Determination (R^2) of 0.93 [25]. Future work includes incorporate socio-economic variables and special events to mitigate errors caused by sudden changes in social activity [25].

Moreover, we observed, accurate electricity demand forecasting is essential for grid stability, economic efficiency, and demand response, particularly in areas experiencing rapid population growth and renewable energy integration [26]. Among the single ML models, there is no universally optimal one. It should be considered that performance depends on variables such as, spatial scale-building, regional, or national, and data availability. In the previous studies, hybrid deep learning architectures have consistently outperformed single ML models and traditional baselines by grasping complementary strengths. CNN-RNN Hybrids excel at capturing local temporal patterns and spatial correlations. Advanced Super-Hybrids achieve the best overall performance, in terms of point forecasting, uncertainty estimation, and interpretability. RNN Variants such as, LSTM, GRU, and their bidirectional versions like, BiLSTM, BiGRU form the essential temporal modeling component. GRU often matches LSTM's accuracy with lower computational cost, while bidirectional variants excel in capturing complex national and peak-demand patterns.

We also observed, at the building level, the importance of forecasting peak demand has increased for demand-side management. LSTM-based models have been applied to large commercial buildings, incorporating operational and contextual features such as historical demand, temperature, time variables, and occupancy indicators. Studies demonstrate that the choice of loss function significantly influences peak demand prediction: while MSE optimizes general accuracy, WMSE performs best for peak-sensitive operations, and NLL provides uncertainty information for risk-aware planning, whereas MAPE tends to underperform [27].

Furthermore, Existing ML models for load forecasting relied on statistical methods like ARIMA, which often struggle with the nonlinear and non-stationary patterns in modern consumption data, with the influence of weather, holidays, and economic activity [28]. In Bangladesh, short and medium-term load forecasting has been explored at regional levels. ML models including Random Forest, Decision Tree, and K-Nearest Neighbor (KNN) have been trained on 30-minute interval data, with Random Forest consistently outperforming others in both short-term (1-hour to 1-week) and medium-term (3-month) horizons [12].

Moving on, deep learning models, particularly CNN-GRU hybrids, have been able to effectively capture both spatial and temporal dependencies across multiple power zones, outperforming GRU, LSTM, and Transformer models in RMSE and MAPE. These hybrid models leverage CNNs for local feature extraction and GRU for efficient sequence learning, enabling accurate week-ahead forecasts, highlighting strong inter-zone correlations and seasonal patterns. However, the model performance shows the need for complete retraining as new observations arrive [29]. And, for long-term electricity demand forecasting, recurrent neural networks such as BiGRU and BiLSTM have demonstrated strong performance in predicting daily peak demand at both national and divisional levels. Using available data that includes demand, generation, temperature, holidays, and maintenance events, BiGRU models achieved a MAPE of approximately 7.3% for forecasts extending

148 days into the future. This approach provides actionable insights for grid operators and highlights the effectiveness of bidirectional recurrent architectures in capturing complex temporal dependencies [26].

Then, Hybrid and super-hybrid models further enhance predictive capability by combining multiple architectures. For instance, Transformer-LSTM-CNN-DNN frameworks integrate the Transformer's attention mechanism for long-range dependencies, LSTM for sequential modeling, CNN for local spatial feature extraction, and DNN for final prediction. Such deeply fused models outperform individual components by effectively capturing both temporal and spatial patterns, though they are computationally intensive and require careful tuning [30]. Similarly, hybrid frameworks that combine LSTM with Prophet models attempt leveraging LSTM's ability to model nonlinear dependencies and Prophet's strength in handling trends, seasonality, and holidays. This hybrid approach consistently outperforms single ML models and traditional statistical approaches like ARIMA [28].

Beyond traditional and hybrid deep learning, explainable and probabilistic forecasting approaches are gaining attention. The deeply fused network (FNET) architecture, combining 1D-CNN and BiLSTM, has been used for point and probabilistic electricity demand forecasting, integrating residual bootstrapping for prediction intervals and SHAP analysis for interpretability. FNET achieved high accuracy, robust trend prediction (88% directional accuracy), and narrow prediction intervals across multiple substations, while identifying key drivers such as previous-day demand, evapotranspiration, and temperature variables. However, challenges still remain in geographic generalizability, data dependency, and computational complexity [31].

Additionally, demand forecasting for specific systems such as air-conditioning units has benefited from hybrid GRU-metaheuristic models. Combining GRU networks with optimization algorithms like Improved Alpine Skiing Optimization (IASO) enables highly accurate next-hour load predictions in multi-zone buildings, achieving R^2 values up to 0.99. These models take building layout, thermal properties, occupancy, weather conditions, and control settings into consideration, though their performance can be affected by unpredictable weather, computational intensity, and limited generalizability to other building types or climates [32].

Now, in optimization studies, a research [22] showed the use of linear optimization framework OSeMOSYS to find the cost-effective generation mixes to meet the power demand. It used national data of existing, and planned generation capacity (coal, gas, heavy fuel oil, hydro, solar, wind, imports) from 2019 to 2045. It made assumptions about fuel prices, and renewable costs. And made demand projections based on public growth estimates. The study took into consideration different scenarios defined by varying fuel prices, renewable cost trajectories, and emissions targets. This methodology gives the least-cost combination of coal, gas, solar, hydro, and imported electricity to meet projected demand. It showed that using ultra-supercritical coal minimizes cost. However, solar PV becomes promising if technology costs are reduced. Finally, this study suggests only focusing on cost minimization without emissions considerations will lead to increased CO₂ emissions by 2045.

Another study by Janefar *et al.* presents a rigorous comparative evaluation of three meta-heuristic algorithms like Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), and Harmony Search (HS) for solving the Economic Load Dispatch (ELD) problem in the context of thermal power plants in Bangladesh[7]. Traditional mathematical methods often fail to address the non-linearities and complexities of contemporary grids, necessitating the use of intelligent optimization [7]. The Economic Load Dispatch (ELD) problem remains a critical optimization challenge in modern power systems, aiming to minimize operational costs while meeting specific power demands . The optimization incorporated complex constraints, including prohibited operating zones, ramp rate limits, and transmission losses modeled via B-coefficients [7]. The results demonstrated that PSO emerged as the most effective algorithm for cost and loss reduction, achieving the lowest power losses (0.71 MW for 6 units and 1.5 MW for 15 units) and generation costs (\$8989.24/h and \$10,550.50/h respectively) [7]. Conversely, while HS resulted in slightly higher costs, it demonstrated superior computational efficiency with the fastest convergence times (3.9 s to 8.0 s) [7]. These findings highlight the trade-offs between speed and optimality, suggesting that PSO is ideal for cost-minimization while HS is suitable for time-sensitive applications [7].

Overall, the reviewed studies highlight key trends in electricity demand forecasting. It can be deduced that hybrid and deeply fused architectures generally outperform standalone models. On the other hand, feature engineering- including environmental, economic, and contextual variables- significantly improves accuracy, and probabilistic or explainable models add operational value by quantifying uncertainty. Despite the advances, there are challenges in extending models across regions and time horizons, scaling them for real-time applications, and integrating high-resolution climate or social-behavioral data. These gaps provide clear directions for future research in accurate and adaptable electricity demand forecasting. Also, the studies suggest that optimization techniques can significantly improve cost-efficient electricity generation planning. Hence, aiding the respective authorities to maintain a sustainable and efficient power system planning in Bangladesh.

2.3 Summary of Key Findings

The key studies from the reviewed literature are summarized in the Table 2.1.

Table 2.1: Key studies from the reviewed literature

Study	Purpose	Key findings	Research gap
[27]	Applying LSTM-based electricity demand forecasting model to a large commercial building complex	Model-MSE: Best for general demand forecasting, but weak for peak-critical decision-making, Model-WMSE: Best for operational peak management and demand response, Model-NLL: Best for risk-aware planning by providing uncertainty information, Model-MAPE: Worst due to unstable learning.	Further tuning and evaluation on extended datasets is recommended. Generalizability to other buildings should be tested with further studies.
[29]	Developing an accurate, reliable, and efficient load forecasting model that predicts week-ahead demand, captures both spatial and temporal dependencies	Among Naive Predictor, and the single models (Transformer, Transformer-LSTM, CNN-Transformer, LSTM, CNN-LSTM, GRU), GRU performed the best, CNN-GRU beats GRU. Among other hybrid models, CNN-GRU outperformed CNN-LSTM, and Transformer-LSTM	The model's performance becomes suboptimal as the forecast period moves further away from the training period.
[26]	Forecasting daily peak and long-term electricity demand nationwide, extending to all eight administrative divisions of Bangladesh	BiGRU best captures the complex temporal dependencies in national peak demand and also provides actionable forecasts for regional grid operators A recursive prediction algorithm, using the BiGRU model, forecasts 148 days into the future, achieving a MAPE of 7.3%	More Transformer-based models need to be explored to enable adaptive, real-time energy systems that work according to demand responses
[12]	Short-Term(1 hour-1 week), Mid-Term (Several weeks/months), and Long-Term (Years/decades) electrical load forecasting in Bangladesh	Random Forest consistently achieves the highest accuracy and lowest error metrics. All models perform slightly worse on the mid-term (3-month) forecast compared to the short-term (1-week) forecast KNN produced the highest error metrics, being the least suitable model	Application of ML models may be done on extended time horizon for long-term forecasting

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Study	Purpose	Key findings	Research gap
[33]	Predicting energy demand in the nine administrative metropolitan areas of Bangladesh	DNN outperforms both LSTM and GRU (close to each other) in terms of MAPE and R^2 in most districts All models perform best in smaller load areas like Sylhet and Barishal, performance decreases in high-demand cities like Dhaka	The DNN model should be applied on other datasets to further validate its effectiveness. Other deep learning models, such as, CNNs, and Transformer-based models should be further explored
[10]	Developing a high-precision forecasting model for electricity consumption in Bangladesh	FB-Prophet is the most reliable and reasonable model for Bangladesh's electricity demand forecasting, with slight pattern deviations in some parts The combination of FB-Prophet with Max Absolute Scaler is identified as the best option for forecasting power consumption	Forecasting using only the historical power demand time-series, without forecasting external variables can be done The proposed pre-processing and feature engineering pipeline can be applied to datasets with different geographical and economic contexts
[30]	Developing a highly accurate short-term (daily) electricity load forecasting model for the nine administrative divisions of Bangladesh	The proposed Transformer–CNN–LSTM–DNN super hybrid model outperforms individual models, overcoming their limitations, and grasping complementary strengths (Transformer's attention + LSTM's sequence memory + CNN's feature detection)	The model can be applied to other datasets to further validate scalability and generalization Alternative CNN architectures and advanced transformer variants can be performed to improve prediction precisions
[32]	Load demand forecasting using GRU/IASO model within air conditioning system	Next-hour forecasting (short-term): Highest R^2 0.99 (explains 99% variance); lowest RMSE, MAE, MSE Next-day forecasting (medium-term): R^2 0.95 (still strong, but slight drop due to longer horizon uncertainty)	The model performance can be extended to other load types Alternative metaheuristics or deep learning architectures can be explored for better efficiency.

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Study	Purpose	Key findings	Research gap
[31]	Introducing the first FNET-based deep fusion model for electricity demand time-series forecasting, combining point forecasting, probabilistic forecasting, and interpretability in a single framework	FNET was the best-performing model across all substations, metrics, and evaluation categories, including point forecasting (highest correlation, lowest RMSE, MAE), stability and error distribution (lowest error variance- STDRE), directional and trend prediction (correctly predicted 88% of demand trends), statistical significance (DM and HLN tests confirmed FNET's improvements over BiLSTM, LSTM-CNN)	The model's effectiveness might heavily rely on the quality and availability of climate data It has been compared to 7 benchmarks, but advanced models were not included The fused CNN-BiLSTM architecture may pose practical deployment challenges, requiring broader testing
[28]	Enhancing short-term electricity demand forecasting accuracy by developing a hybrid model, combining LSTM networks and the Prophet model	The Hybrid Model achieves the best scores across all three key metrics: Lowest RMSE (65.34), Lowest MAPE (7.3%), Highest R^2 (0.98) Contrary to the Prophet model, LSTM is confirmed as the best standalone model. The hybrid model significantly outperforms LSTM	The hybrid framework can be extended by incorporating attention mechanisms, Graph Neural Networks (GNNs). The model should be evaluated across diverse datasets from different geographical regions and energy markets.
[13]	Introducing a hybrid DL model, combining CNNs and stacked Bidirectional LSTM for short-term electricity demand forecasting in Dhaka City	The proposed model achieved a MAPE of 1.64%. When evaluated against benchmark models (LSTM, CNN-LSTM, and CNN-BiLSTM), it consistently achieved the lowest MSE, RMSE, MAE, and MAPE	The study focus only on Dhaka's demand patterns, without considering wider grid interactions

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Study	Purpose	Key findings	Research gap
[23]	Applying ML models (KNN, ANN, and GBRT) to predict power demand and fuel consumption in a 150 MW HFO power plant in Bangladesh	KNN: Validation MSE increased by 27% in the 8:2 split ANN: Achieved very high R^2 values (0.998) across all splits. Faster convergence in the 8:2 split indicates quicker learning GBRT: Consistently delivered the highest accuracy and stability, achieving near-perfect R^2 values (0.999) across training, validation, and test sets	The model is specific to one plant's operational profile and may not generalize to the full national grid or different plant types
[14]	Introducing the Time Series Dense Encoder (TiDE) model, a DL architecture previously unused in this specific regional context	When compared against Linear Regression, XGBoost, and an Ensemble model, TiDE demonstrated superior accuracy across 7, 15, and 30-day forecast horizons	The study focuses exclusively on a single region, needs to be tested on broader area and different datasets
[15]	To develop a reliable ML forecasting system that can predict electricity demand in Dhaka city using historical load, temperature, and seasonal patterns	LightGBM was the best-performing model, with lowest RMSE, MAE, and the only model with a positive R^2 (0.03) SARIMA performed better than most ML models but failed to capture temperature and holiday effects, getting a negative R^2 (-0.11) LSTM performed the worst (RMSE 3997, R^2 -23.36) due to insufficient data size and high noise	Different sources of energy production such as solar energy, biogas, microgrid should be worked with to forecast the load effectively.
[22]	Find the cost-effective generation mixes to meet the power demand	Using ultra-supercritical coal minimizes cost. However, solar PV becomes promising if technology costs are reduced.	Relies on projected demand and fuel price assumptions.

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Study	Purpose	Key findings	Research gap
[7]	Solve the Economic Load Dispatch problem for thermal plants in Bangladesh.	PSO emerged as the most effective algorithm for cost and loss reduction, achieving the lowest power losses (0.71 MW for 6 units and 1.5 MW for 15 units) and generation costs (\$8989.24/h and \$10,550.50/h respectively).	The study is limited to thermal power plants. Also renewable energy sources (solar/wind) and more complex hybrid metaheuristic algorithms should be integrated.

Chapter 3

Requirements, Impacts and Constraints

3.1 Final Specifications and Requirements

This project develops an integrated framework for hourly electricity-demand forecasting and generation-dispatch optimization in Bangladesh. The framework consists of two sequential components: (i) electricity-demand forecasting and (ii) cost-oriented generation allocation.

3.1.1 Functional Requirements

The system is required to fulfill the following functions:

1. Bangladesh's national hourly electricity demand using historical records from the Power Grid Bangladesh PLC (PGCB), weather-related variables, and selected macroeconomic features. Forecasting performance is evaluated using Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).
2. Given historical or forecast demand, determine the generation allocation across ten aggregated electricity-source categories while minimizing the adopted generation-cost function. The optimization must satisfy power-balance constraints, source-specific capacity limits, and availability restrictions.
3. Implement and compare multiple optimization approaches, including an exact mathematical optimization method, metaheuristic algorithms, and a learned optimization method based on a Learned Augmented Neural Network (LANN).
4. Use the reference optimization formulation to evaluate the quality of alternative solutions. The reference solution is treated as optimal under the adopted mathematical formulation and assumptions, subject to the validity of the solver and verification procedures.
5. Estimate potential reductions in variable generation cost under the adopted assumptions. Reported reductions must be interpreted as model-estimated results rather than guaranteed savings in real-world power-system operation.

3.1.2 Data Requirements

The project uses publicly available and aggregated data sources. The principal datasets include:

- PGCB historical electricity records: Hourly generation and demand data covering 2015–2025 [11].
- Open-Meteo historical weather data: Weather variables used as potential explanatory features for demand forecasting.
- World Bank indicators: Population, gross domestic product (GDP), and foreign direct investment (FDI) data.
- BPDB Daily Electricity Generation Reports: 178 reports covering January–June 2025, used to extract source-level generation, capacity, and availability information [34].
- BPDB Annual Report 2024–25: Used as a supplementary source for generation-related cost and system information [35].

3.1.3 Software and Hardware Requirements

The implementation is developed in Python using Google Colab. The principal software libraries include:

Category	Tools and Libraries
Data processing	pandas, NumPy
Machine learning	scikit-learn, XGBoost, LightGBM
Deep learning	TensorFlow/Keras, PyTorch
Hyperparameter tuning	Optuna
Numerical optimization	SciPy, HiGHS, SLSQP
PDF extraction	PyMuPDF
Visualization	Matplotlib, Seaborn

Table 3.1: Software library requirements

The forecasting models and LANN-based optimization experiments need to use a Google Colab GPU. Classical machine-learning models, linear or quadratic programming, and metaheuristic optimization methods do not require specialized hardware beyond a conventional computational environment.

3.1.4 Operational Scope

The proposed framework is intended for research-based decision support, not direct autonomous control of the national electricity grid. The optimization model operates on aggregated generation categories rather than individual generating units and does not represent every operational feature of a real power system. The current implementation excludes unit commitment, startup/shutdown and ramping costs, transmission constraints, detailed reserves, contractual/capacity payments, capital and fixed operating costs, real-time dispatch, and explicit carbon constraints. These limitations should be considered when interpreting the optimization results.

3.2 Societal Impact

Accurate demand forecasting can support generation planning by reducing mismatches between expected demand and available supply. Under-forecasting may contribute to inadequate generation planning and load-shedding, while over-forecasting may result in inefficient allocation of generation resources.

The dispatch model indicates a potential reduction in variable generation cost under the adopted assumptions. Hence, the estimated cost reduction should not be interpreted as an audited or guaranteed saving. Moreover, the study does not directly assess load shedding, regional energy access, employment effects, fuel usage, cost changes.

3.3 Environmental Impact

The dispatch model minimizes generation cost without explicitly considering emissions, carbon pricing, or environmental variables. Under the adopted assumptions, the optimized dispatch increases gas and coal generation shares, while reducing HFO/liquid-fuel generation. Although this reduces modeled generation cost, it does not necessarily reduce environmental impact, particularly because coal has higher carbon intensity than gas. The results should therefore be interpreted as system-level economic and operational insights rather than a complete assessment of energy equity or sustainability.

3.4 Ethical Issues

The study uses only aggregate, publicly available data and does not involve personal information. Results are reported transparently by distinguishing official, derived, and assumed inputs, and the estimated cost reduction is presented as a model-based result rather than a guaranteed real-world outcome. The framework is intended for planning and decision support, not autonomous grid control. Any practical deployment would require appropriate engineering and regulatory review. All external datasets and sources are identified and cited according to their applicable usage and attribution requirements.

Chapter 4

Proposed Methodology

4.1 Methodology Overview

The workflow of this study is shown in Figure 4.1. The hourly data for electricity demand in Bangladesh, from 2015 to 2025, were collected from the PGCB website [11]. During data cleaning, duplicate values were removed and missing values were imputed using linear interpolation to maintain time-series continuity. To reduce the impact of extreme values in the target column, percentile-based outlier detection was applied. The 1st and 99th percentile values were used as lower and upper bounds, respectively. Any value below the 1st percentile was clipped to the lower limit, and any value above the 99th percentile was clipped to the upper limit.

The experiments in this study were conducted in two settings: (i) using the complete dataset and (ii) using rolling-window subsets of the dataset. In the rolling-window approach, the dataset was divided into multiple subsets, each covering a four-year interval. Each subset was further split into training, validation, and testing sets using a 2:1:1 year ratio.

Several variables were removed during preprocessing to achieve a realistic forecasting setting and to minimize potential information leakage. In particular, the Generation (MW) feature was excluded, as electricity generation is typically adjusted in response to demand and may not be available at prediction time. Including this variable could therefore introduce target leakage and lead to overly optimistic model performance. Similarly, generation composition variables such as gas, liquid fuel, coal, hydro, solar, and wind were excluded, as they primarily represent supply-side characteristics rather than direct determinants of electricity demand. In addition, electricity import source variables (India Bheramara HVDC, India Tripura, India Adani, and Nepal) were excluded, as electricity demand is generally independent of specific import sources. Therefore, the study focuses on historical demand patterns and temporal features to enhance model generalization.

In addition to the features provided in the dataset, relevant external features were incorporated to achieve better forecasting performance. These included holidays, seasonal indicators, socio-economic variables (GDP, population, and FDI), and shock-event indicators such as national grid failures and nationwide lockdowns.

The study first evaluates traditional machine learning models using engineered temporal features, followed by deep learning architectures that learn temporal dependencies directly from sequential data.

For machine learning models, time-based features such as hour, day of the week, and month were extracted from the datetime index. To capture cyclical patterns, sine and

cosine transformations were applied to the hour and month features. Lag features were introduced to model temporal dependencies, including demand from the previous hour, the same hour of the previous day, and the same hour of the previous week. A rolling feature representing the previous day's mean demand was also included to capture short-term trends. Standard scaling was applied to feature columns, with parameters fitted on the training set and applied to validation and testing sets. Traditional machine learning models—including linear regression, ridge regression, lasso regression, random forest regression, gradient boosting regression, XGBoost regression, and LightGBM regression—were trained using the training dataset and evaluated on the validation dataset.

For deep learning-based forecasting, sequential input data were constructed from the target variable using fixed-length time windows. Unlike traditional models, manual feature engineering (such as lag and rolling features) was not applied, as deep learning models can learn temporal dependencies directly from sequential patterns. Min-Max scaling was applied to normalize the input data. Deep learning models, including LSTM, GRU, Bi-LSTM, and Bi-GRU, were trained using the sequential dataset.

Hyperparameters were tuned to minimize prediction error, and models were evaluated repeatedly on the validation dataset to monitor overfitting. Finally, the tuned models were used to predict outcomes on the test dataset. Model performance was evaluated using standard error metrics, including Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). A comparative analysis was conducted across all models to identify the best-performing approach. The overall workflow of the study, from raw data collection through preprocessing, model training, and evaluation, is illustrated in Figure 4.1.

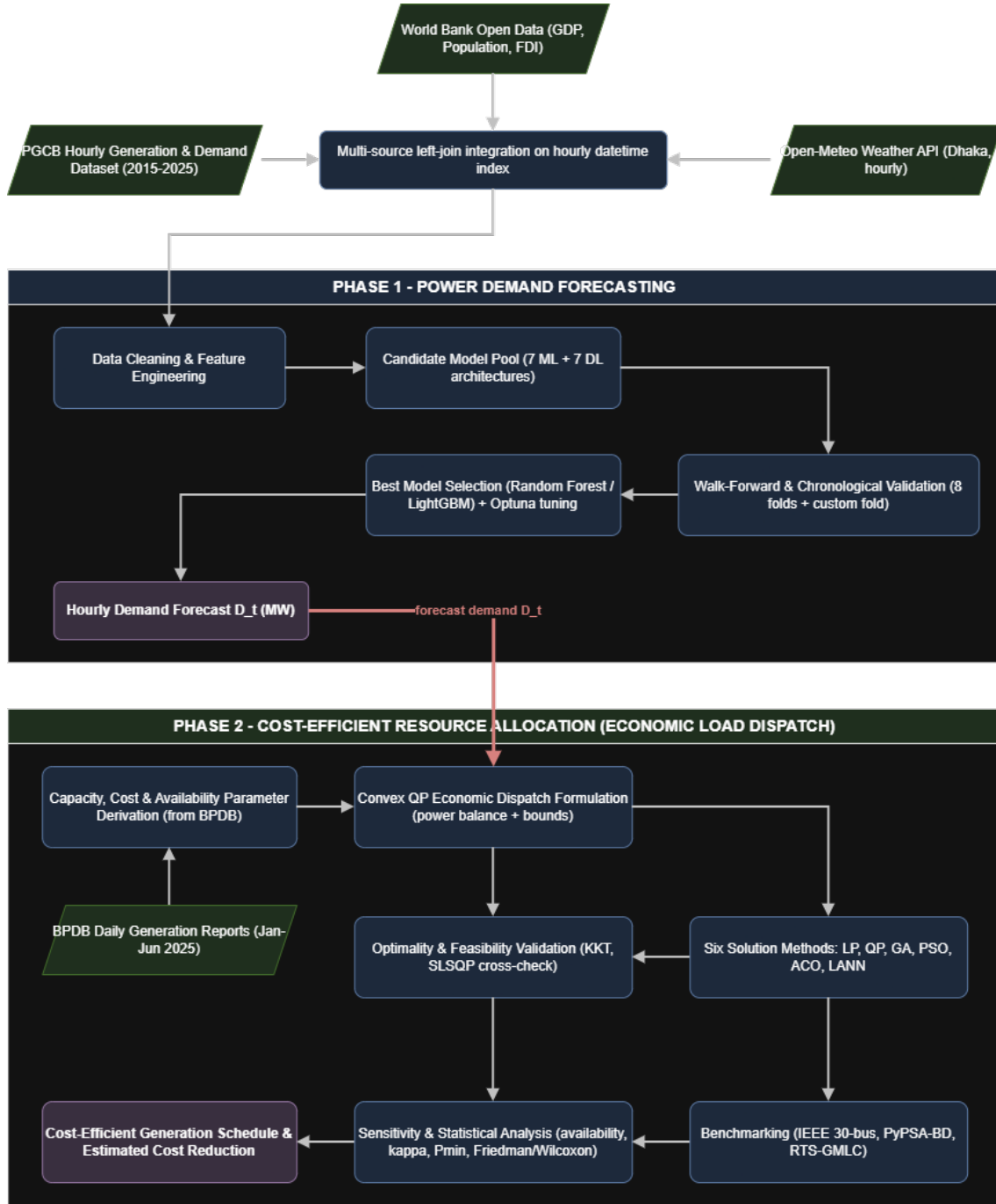


Figure 4.1: Workflow Diagram of the Study

4.2 Forecasting Dataset Collection and Description

This thesis uses the PGCB Hourly Generation Dataset [11] as the primary data source. This dataset was collected from the UCI Machine Learning Repository [36], which originally sourced it from the National Power Grid Company of Bangladesh's website [11]. Along with this primary source, the study utilizes the Open-Meteo Archive API [37] to collect meteorological data, and the World Bank Open Data [38] repository to integrate macroeconomic impacts on electricity demand in Bangladesh. The rationale behind integrating these factors into the dataset is to assist models in capturing the non-linear interac-

tions among time, climate, economy, and socio-cultural complexity in Bangladesh’s electricity demand [39, 40]. A sample of the core columns from the raw PGCB dataset—including the datetime index, demand, generation, load shedding, and remarks fields—is shown in Table 4.1.

Table 4.1: PGCB Dataset Sample: Core Columns

Datetime	Generation (MW)	Demand (MW)	Load Shedding	Remarks
2015-04-19 22:00	6323.00	6323	0	–
2015-04-19 21:00	6667.00	6667	0	–
2015-04-19 19:00	6897.00	6897	0	–
2015-04-19 18:30	6933.00	6933	0	Evening Peak
2015-04-19 18:00	6874.00	6874	0	–

The domestic generation breakdown from the raw PGCB dataset is illustrated in Table 4.2.

Table 4.2: PGCB Dataset Sample: Domestic Generation (MW)

Datetime	Gas	Liquid Fuel	Coal	Hydro	Solar	Wind
2015-04-19 22:00	–	–	–	–	–	–
2015-04-19 21:00	–	–	–	–	–	–
2015-04-19 19:00	4415	1836	161	41	–	–
2015-04-19 18:30	4423	1862	159	45	–	–
2015-04-19 18:00	4319	1892	155	65	–	–

The cross-border electricity import columns from the raw PGCB dataset are shown in Table 4.3.

Table 4.3: PGCB Dataset Sample: Imports (MW)

Datetime	India (HVDC)	India (Tripura)	India (Adani)	Nepal
2015-04-19 22:00	0	0	–	–
2015-04-19 21:00	0	0	–	–
2015-04-19 19:00	444	0	–	–
2015-04-19 18:30	444	0	–	–
2015-04-19 18:00	443	0	–	–

Primary Dataset: PGCB Hourly Generation and Demand Records

The primary dataset from PGCB was collected as a CSV file and uploaded to Google Drive to enhance reproducibility, eliminating manual data handling by reading the link directly using Python Pandas’ `read_csv()` function. The dataset contains roughly **92,000+** hourly observations along with 15 features spanning from January **2015** to June **2025**, covering nearly a decade of national power grid supply and demand data. Table 4.4 summarizes all original PGCB dataset fields, their descriptions, and their respective treatment in the preprocessing pipeline.

Table 4.4: Original PGCB dataset fields, their descriptions, and their treatment in the preprocessing pipeline.

Field	Description	Role in Model
datetime	Timestamp (sub-hourly intervals, resampled to 1h)	Index / temporal anchor
demand_mw	System demand in megawatts	Primary target variable
generation_mw	Total generation dispatched (MW)	Dropped after target formulation; supply-side feature, risk of leakage
load_shedding	Duration/magnitude of loadshedding per interval	Conditionally dropped (< 30,000 non-zero rows threshold)
gas, liquid_fuel, coal, hydro, solar, wind	Fuel-type generation disaggregation (MW)	Dropped — supply-side features; target leakage risk [39]
india_bheramara_hvdc, india_tripura, india_adani, nepal	Cross-border import volumes (MW)	Dropped — supply-side constraints, not consumer demand drivers
remarks	Peak type annotation (e.g., 'morning peak', 'evening peak')	One-hot encoded into peak_type_* binary features (Step 7)

Weather Data: Open-Meteo Historical Archive API

Meteorological data were integrated via REST API calls from Open-Meteo [37], which provides detailed historical weather records for geographical locations using coordinates at no cost. Incorporating weather data—specifically temperature, air density, and atmospheric pressure—was motivated by the goal of reducing MAPE relative to baseline models [39, 41]. Weather data were retrieved for Latitude 23.81° N, Longitude 90.41° E (Dhaka metropolitan area), as it represents the largest urban center driving national electrical load patterns in Bangladesh.

Macroeconomic Data: World Bank Open Data API

Annual macroeconomic data were integrated into the DataFrame from World Bank Open Data [38] using the Python `pandas_datareader` library via the `wb.download()` endpoint API call. A direct connection exists between macro-level economic variables (such as GDP and FDI) and overall electricity demand. Following a framework similar to Islam et al. [39], three indicators—total population (`SP.POP.TOTL`), gross domestic product in current USD (`NY.GDP.MKTP.CD`), and net foreign direct investment inflows (`BX.KLT.DINV.CD.WD`)—were collected for the period 2015–2025.

Merging Data: A left-join approach was used to consolidate all four data sources into a single modeling DataFrame, using the hourly datetime index as the common key. First, the PGCB CSV was loaded, parsed, and set as the datetime index. Second, weather data (wind speed, precipitation) retrieved for the Dhaka metropolitan coordinates were merged on exact hourly timestamps. Third, annual macroeconomic data from the World Bank were expanded to hourly resolution by duplicating annual values across all hourly records within each corresponding year, followed by a join using a year-level key. Fourth, public holiday binary flags were generated using the `holidays` Python library and merged directly. All joins were left-joins anchored on the primary PGCB time index, ensuring no core observations were lost; external data gaps were filled with NaN values and addressed during missing-value imputation. This multi-source feature fusion approach mirrors methods applied in recent electricity demand forecasting literature [39, 40].

4.3 Forecasting Dataset Preprocessing

4.3.1 Data Cleaning

The data cleaning process followed a multi-stage strategy.

The initial processing step normalized the dataset’s time index for time-series modeling. The primary datetime column was auto-detected by scanning field names for keywords (`datetime`, `date`, `time`), parsed to Pandas Timestamp objects via `pd.to_datetime()`, and established as the DataFrame index. Redundant timestamps were removed by retaining only the first instance. Power grid collection systems frequently introduce duplicate entries due to Daylight Saving Time (DST) adjustments or manual data entry errors.

Redundant Column Removal: According to established literature, electricity demand records in Bangladesh often reflect grid supply capacity rather than true unconstrained consumer demand [39, 42]. Consequently, to prevent target leakage, all supply-side generation features—including fuel-type breakdowns (`gas`, `liquid_fuel`, `coal`, `hydro`, `solar`, `wind`) and cross-border imports (India Bheramara HVDC, India Tripura, India Adani, Nepal)—were excluded from the feature set. Without this removal, models might learn identity constraints between produced supply and demand rather than underlying human consumption dynamics. The `generation_mw` column was used solely during data

cleaning diagnostics and was excluded from model training.

To ensure temporal continuity, missing timestamps were addressed. Sub-hourly records were resampled to a uniform hourly frequency using mean and maximum aggregation for continuous variables, alongside mode-based aggregation for the categorical remarks column. Missing values in continuous variables (generation_mw and demand_mw) were imputed using bi-directional linear interpolation, followed by forward and backward filling to remove any remaining NaN values.

Load Shedding Column Removal: The load_shedding column contained a high proportion of zeros, providing minimal feature variance. A conditional threshold was implemented: if non-zero values were present in fewer than 30,000 hours, the entire column was dropped.

4.3.2 Data Transformation

Time-based features require cyclical encoding rather than linear scaling to allow predictive models to properly interpret periodic boundary transitions (e.g., the proximity between 23:00 and 00:00). As demonstrated by Al-Amin [43], cyclic temporal features should be projected onto a unit circle using sine and cosine trigonometric transformations: $\sin\left(\frac{2\pi \cdot \text{value}}{\text{period}}\right)$ and $\cos\left(\frac{2\pi \cdot \text{value}}{\text{period}}\right)$. A similar categorical mapping was applied to seasonal data (Winter=0, Summer=1, Monsoon=2, Post-Monsoon=3). The complete trigonometric encoding specifications are detailed in Table 4.5.

One-Hot Encoding: The remarks column in the primary dataset categorizes peak periods into day_peak, evening_peak, and unannotated periods. One-hot encoding was applied to create three binary indicator variables, enabling models to isolate demand phases across daily cycles.

Feature Scaling: Features were normalized using RobustScaler. To prevent data leakage, the scaler was fitted exclusively on training sets and applied across training, validation, and test splits. Standard scaling tools (such as MinMaxScaler or StandardScaler) can be overly sensitive to demand spikes and macroeconomic outliers. RobustScaler mitigates this by centering features around their median and scaling based on the Interquartile Range (IQR):

$$x_{\text{scaled}} = \frac{x - \text{median}_{\text{train}}}{\text{IQR}_{\text{train}}} \quad (4.1)$$

Table 4.5: Trigonometric cyclical encoding of temporal features applied to all hourly observations.

Feature	Formula	Captures
season_sin / season_cos	$\sin(2\pi \cdot \text{season}/4), \cos(2\pi \cdot \text{season}/4)$	Bangladesh’s 4-season cycle (winter, summer, monsoon, post-monsoon)
month_sin / month_cos	$\sin(2\pi \cdot \text{month}/12), \cos(2\pi \cdot \text{month}/12)$	Annual intra-year thermal load cycle
hour_sin / hour_cos	$\sin(2\pi \cdot \text{hour_decimal}/24), \cos(2\pi \cdot \text{hour_decimal}/24)$	Diurnal load profile (morning peak, evening peak, nocturnal trough)

4.3.3 Data Integration

External datasets were merged with the primary PGCB temporal index using left-join operations.

Shock Features: When incorporating weather data from Open-Meteo, shock indicators were conditionally derived. A binary variable, `is_weather_shock`, was set to 1 if wind speed (> 40 km/h) and precipitation (> 15 mm/h) exceeded thresholds based on Bangladesh Meteorological Department criteria for severe weather in the Bengal delta region. Additionally, 15 major historical events (e.g., cyclone landfalls, national elections, COVID-19 lockdowns, civil unrest) were encoded as a binary variable, `is_manual_shock`. Combining both yields the final composite indicator: `is_shock`.

Holidays Integration: A calendar variable, `days_of_week`, was extracted using the Python `holidays` library, with Fridays and Saturdays designated as weekends in the binary variable `is_holiday`. Holiday indicators significantly enhance demand forecasting accuracy, as highlighted by Ferdaus et al. [40].

Economic Indicators: Hourly-broadcasted macroeconomic variables from the World Bank were mapped to the dataset. Derived indicators—GDP per capita (in thousands of USD) and FDI as a percentage of GDP—were calculated as follows:

$$\text{GDP_per_capita} = \frac{\text{GDP} \times 10^9}{(\text{Population} \times 10^6) \times 1000} \quad (4.2)$$

$$\text{FDI_pct_GDP} = \left(\frac{\text{FDI}}{\text{GDP}} \right) \times 100 \quad (4.3)$$

Lag and Rolling Features: Autoregressive lag and rolling features were incorporated to capture autocorrelation patterns in electricity demand. In alignment with walk-forward validation best practices, lag and rolling features were calculated on the entire dataset prior to splitting into train, validation, and test partitions [44]. This prevents boundary NaN issues at partition edges. Figure 5.13 illustrates the Pearson correlation coefficients between lag/rolling features and target demand.

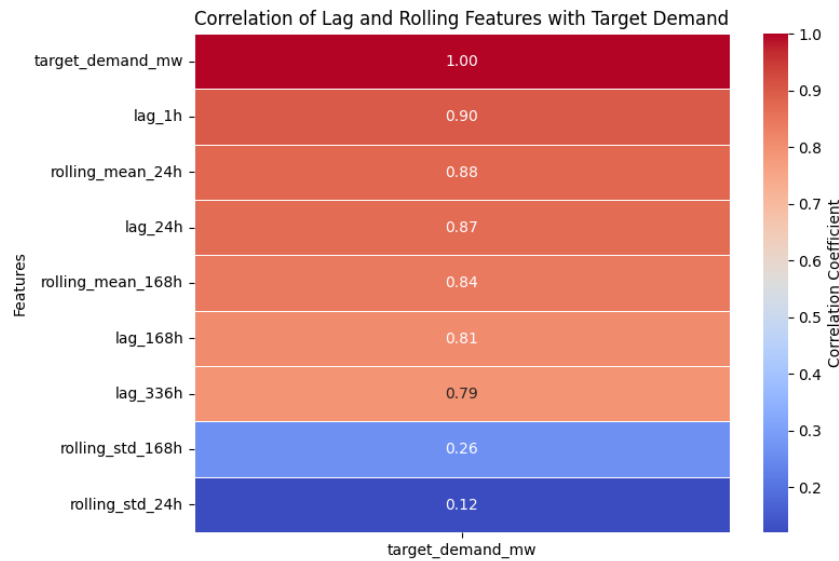


Figure 4.2: Correlation of Lag and Rolling Features with Target Demand

The final feature set includes the following autoregressive attributes:

- `lag_1h`: `demand_mw` lagged by 1 hour (immediately preceding observation).
- `lag_24h`: `demand_mw` lagged by 24 hours (same hour on the previous day).
- `lag_168h`: `demand_mw` lagged by 168 hours (same hour on the previous week).
- `lag_336h`: `demand_mw` lagged by 336 hours (same hour two weeks prior).
- `rolling_mean_24h`: 24-hour moving average of `lag_1h`.
- `rolling_mean_168h`: 168-hour moving average of `lag_1h`.
- `rolling_std_24h`: 24-hour moving standard deviation of `lag_1h`.
- `rolling_std_168h`: 168-hour moving standard deviation of `lag_1h`.

All rolling features were computed strictly on shifted target values (`lag_1h`) to eliminate look-ahead bias and prevent temporal leakage [45]. The initial 336 observations were used for lag initialization and excluded from modeling. The composition of the integrated feature matrix is summarized in Table 4.6.

Table 4.6: Final integrated feature matrix composition after all collection, cleaning, transformation, and integration steps.

Feature Group	Count	Features Included
Temporal / Calendar	9	<code>hour_sin</code> , <code>hour_cos</code> , <code>month_sin</code> , <code>month_cos</code> , <code>season_sin</code> , <code>season_cos</code> , <code>day_of_week</code> , <code>is_weekend</code> , <code>is_holiday</code>
Shock / Event Flags	1	<code>is_shock_event</code> (composite: <code>weather_threshold</code> OR manual shock dates)
Peak-Type Dummies	4–6	<code>peak_type_Morning Peak</code> , <code>peak_type_Evening Peak</code> , <code>peak_type_Night</code> , <code>peak_type_None</code> , etc.
Lag Features	4–5	<code>lag_1h</code> , <code>lag_24h</code> , <code>lag_168h</code> , <code>lag_336h</code> [+ <code>lag_8760h</code> in walk-forward]
Rolling Statistics	4	<code>rolling_mean_24h</code> , <code>rolling_mean_168h</code> , <code>rolling_std_24h</code> , <code>rolling_std_168h</code>
Macroeconomic	4–5	<code>population</code> , <code>gdp</code> , <code>fdi</code> , <code>gdp_per_capita</code> , <code>fdi_pct_gdp</code> [or <code>yoy_growth_30d</code> in walk-forward]
Target Variable	1	<code>target_demand_mw</code> (= <code>demand_mw</code>)

4.3.4 Data Reduction

Redundant variables and potential leakage sources (e.g., `holiday_name`, `year`, `hour_decimal`) were removed. Chronological partitioning was applied to construct train, validation, and test sets for both the base pipeline and walk-forward evaluations. For the base pipeline, approximately 80% of historical data (up to end-of-year 2022) were allocated to training while preserving chronological order. The partition details are presented in Table 4.7.

Table 4.7: Chronological train-validation-test partition used in the base pipeline.

Split	Years	Approx. Records	Purpose
Training	2015–2022	~70,000 hrs	Model fitting and parameter learning
Validation	2023	~8,760 hrs	Hyperparameter selection, early stopping
Test	2024–2025	~12,000+ hrs	Final unbiased out-of-sample evaluation

In the walk-forward configuration, sequential 2-year windows were allocated for training, followed by 1 year for validation and 1 year for testing. Additionally, a custom fold was evaluated using 4 training years, 1 validation year, and 1 test year. Unlike standard K -fold cross-validation, which shuffles observations and risks leaking future context into past windows, temporal walk-forward evaluation ensures mathematical validity for time-series forecasting [44, 45, 46].

4.3.5 Summary of Preprocessed Data

The final modeling dataset is a leak-free hourly time series spanning from April 2015 to June 2025. Key dataset characteristics are summarized in Table 4.8.

Table 4.8: Summary of Final Preprocessed Dataset

Property	Value
Raw dataset time range	April 2015 – June 2025
Raw observations (sub-hourly)	~92,000+
After hourly resampling	~87,600 rows
After lag warm-up removal (336 h)	~87,264 rows
Training set (base pipeline)	2015–2022 (~70,000 rows)
Validation set (base pipeline)	2023 (~8,760 rows)
Test set (base pipeline)	2024–2025 (~12,000 rows)
Total engineered features	26 (see Table 4.9)
Target variable	target_demand_mw (MW)
Residual missing values	0

The engineered feature matrix is cataloged by functional category in Table 4.9.

Table 4.9: Complete Feature List — Category, Name, and Description

Category	Feature Name	Description
Temporal / Calendar	hour_sin	Sine encoding of decimal hour-of-day (period = 24 h)
Temporal / Calendar	hour_cos	Cosine encoding of decimal hour-of-day (period = 24 h)

Continued on next page

Table 4.9: Complete Feature List (continued)

Category	Feature Name	Description
Temporal / Calendar	month_sin	Sine encoding of calendar month (period = 12)
Temporal / Calendar	month_cos	Cosine encoding of calendar month (period = 12)
Temporal / Calendar	season_sin	Sine encoding of 4-category season index (period = 4)
Temporal / Calendar	season_cos	Cosine encoding of 4-category season index (period = 4)
Temporal / Calendar	day_of_week	Integer day index (0 = Monday ... 6 = Sunday)
Temporal / Calendar	is_weekend	Binary flag: 1 for Friday/Saturday (BD weekend), 0 otherwise
Temporal / Calendar	is_holiday	Binary flag: 1 if date is a Bangladesh public holiday
Shock / Event	is_shock_event	Composite binary: 1 if weather shock OR listed manual shock date
Peak-Type Dummies	peak_type_Day_Peak	OHE: 1 if PGCB remarks column records a Day Peak observation
Peak-Type Dummies	peak_type_Evening_Peak	OHE: 1 if PGCB remarks column records an Evening Peak observation
Peak-Type Dummies	peak_type_None	OHE: 1 if no peak-type remark was recorded
Lag Features	lag_1h	Demand (MW) at $t - 1$ h (immediately preceding hour)
Lag Features	lag_24h	Demand (MW) at $t - 24$ h (same hour, previous day)
Lag Features	lag_168h	Demand (MW) at $t - 168$ h (same hour, previous week)
Lag Features	lag_336h	Demand (MW) at $t - 336$ h (same hour, two weeks prior)
Rolling Statistics	rolling_mean_24h	Mean of lag_1h over the preceding 24 h window
Rolling Statistics	rolling_mean_168h	Mean of lag_1h over the preceding 168 h window
Rolling Statistics	rolling_std_24h	Std. dev. of lag_1h over the preceding 24 h window
Rolling Statistics	rolling_std_168h	Std. dev. of lag_1h over the preceding 168 h window
Macroeconomic	population	Bangladesh total population (World Bank SP.POP.TOTL), annual
Macroeconomic	gdp	Bangladesh GDP in current USD (World Bank NY.GDP.MKTP.CD), annual

Continued on next page

Table 4.9: Complete Feature List (continued)

Category	Feature Name	Description
Macroeconomic	fdi	Net FDI inflows USD (World Bank BX.KLT.DINV.CD.WD), annual
Macroeconomic	gdp_per_capita	Derived: GDP / Population (thousands of USD)
Macroeconomic	fdi_pct_gdp	Derived: FDI as a percentage of GDP

All autoregressive lag and rolling variables were generated on the full dataset prior to partitioning. RobustScaler parameters were learned purely from training subsets. Figure 4.3 presents the Spearman rank correlations between features and target demand.

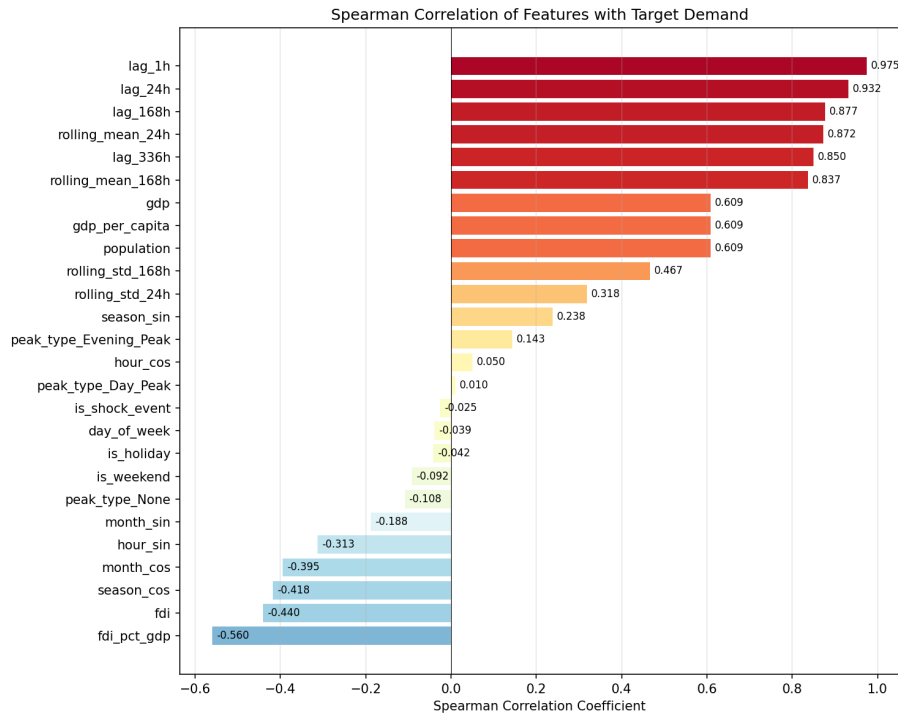


Figure 4.3: Spearman rank correlation between each engineered feature and the target demand.mw. Lag and rolling features dominate the top ranks, confirming the strong autoregressive structure of the demand series.

4.4 Forecasting Models and Experimental Design

To evaluate forecasting methodologies for Bangladesh’s electrical grid, a diverse set of predictive models was benchmarked, spanning classical machine learning models and deep learning sequence architectures.

4.4.1 Models Evaluated

- **Linear Regression:** Standard baseline linear model to evaluate linear feature relationships.

- **Ridge Regression:** Linear regression with an L_2 regularization penalty ($\alpha = 1.0$) to mitigate potential multicollinearity among correlated historical lags.
- **Lasso Regression:** Linear model incorporating an L_1 penalty ($\alpha = 0.001$, `max_iter = 5000`) for embedded feature selection.
- **Random Forest Regressor:** Bagging ensemble of decision trees (`n_estimators = 100`, `max_depth = 8`, `min_samples_split = 20`, `min_samples_leaf = 10`, `random_state = 42`).
- **Gradient Boosting Regressor:** Sequential boosting ensemble (`n_estimators = 150`, `learning_rate = 0.05`, `max_depth = 5`, `random_state = 42`).
- **XGBoost:** Optimized gradient boosted decision tree model (`n_estimators = 1000`, `learning_rate = 0.05`, `max_depth = 5`, `reg_alpha = 0.1`, `reg_lambda = 1.0`, `early_stopping_rounds = 50`, `eval_metric = MAPE`).
- **LightGBM:** Efficient leaf-wise gradient boosting architecture (`n_estimators = 1000`, `learning_rate = 0.05`, `max_depth = 5`, `reg_alpha = 0.1`, `reg_lambda = 1.0`).
- **LSTM:** Long Short-Term Memory recurrent neural network designed to capture long-term temporal dependencies.
- **BiLSTM:** Bidirectional LSTM network evaluating contextual sequence information simultaneously in forward and backward temporal directions.
- **GRU:** Gated Recurrent Unit neural network featuring simplified gating structures for faster convergence.
- **BiGRU:** Bidirectional GRU network combining bidirectional temporal context with lower parameter overhead.
- **TCN:** Temporal Convolutional Network utilizing 1D causal dilated convolutions for non-recurrent sequence modeling.
- **CNN-BiLSTM:** Hybrid neural network applying 1D convolutional feature extractors followed by bidirectional LSTM layers.
- **Attention-LSTM:** LSTM architecture equipped with soft-attention mechanisms to dynamically weight critical temporal lookbacks.

4.4.2 Validation Strategy

Walk-Forward Validation

To respect temporal ordering, an 8-fold walk-forward validation strategy was implemented. Models were tested exclusively on historical observations chronological succeeding their training windows. Table 4.10 details window allocations across folds.

Implementation Notes:

- Scaling parameters (`RobustScaler`) were learned purely from training windows within each fold.

Table 4.10: Walk-Forward Validation Setup Across Folds

Fold	Training Window	Validation Window	Testing Window
1	2015–2016	2017	2018
2	2016–2017	2018	2019
3	2017–2018	2019	2020
4	2018–2019	2020	2021
5	2019–2020	2021	2022
6	2020–2021	2022	2023
7	2021–2022	2023	2024
8	2022–2023	2024	2025

- Early stopping was applied during XGBoost and LightGBM training to avoid over-fitting on validation splits.
- **Fold 1** was excluded from overall ranking aggregations due to extreme historical distribution shifts during that period.

4.4.3 Development Environment

Experiments were executed in Python 3 on Google Colaboratory runtimes with GPU acceleration. Core software libraries are detailed in Table 4.11.

Table 4.11: Software Libraries and Frameworks

Library / Framework	Version	Purpose
pandas	≥ 2.0	Data ingestion, resampling, feature engineering
NumPy	≥ 1.24	Numerical array operations
scikit-learn	≥ 1.3	Preprocessing (RobustScaler), linear models, RF, GB, metrics
XGBoost	≥ 2.0	Gradient-boosted decision tree regressor
LightGBM	≥ 4.0	Histogram-based gradient boosting regressor
TensorFlow / Keras	≥ 2.15	Deep learning model construction and training (GPU)
Optuna	≥ 3.5	Hyperparameter optimization (TPE sampler)
openmeteo-requests	≥ 0.2	Open-Meteo Historical Weather API client
pandas-datareader	≥ 0.10	World Bank Open Data API (wb.download)
holidays	≥ 0.46	Bangladesh public holiday calendar generation
matplotlib / seaborn	≥ 3.8	Visualization and model comparison charts

4.4.4 Preprocessing and Feature Engineering Pipeline

The data preparation pipeline consists of twelve sequential steps:

Step 1 — Dataset loading: Primary CSV files are loaded, timestamps converted to Pandas Timestamp objects, set as index, and ordered chronologically.

Step 2 — Column removal: Supply-side and cross-border import columns are dropped. `generation_mw` is saved strictly for diagnostic cleaning.

Step 3 — Deduplication and resampling: Duplicate timestamps are dropped. Sub-hourly data are resampled to hourly frequency. `load_shedding` is removed if non-zero entries are under 30,000 hours.

Step 4 — Calendar and shock generation: Meteorological attributes are retrieved from Open-Meteo. Binary shock flags (`is_weather_shock`, `is_manual_shock`) form the composite `is_shock_event` attribute. Holiday flags are generated via `holidays`.

Step 5 — Seasonal encoding: Cyclic sine/cosine transformations encode season, month, and hour attributes.

Step 6 — Macroeconomic integration: Annual World Bank variables are expanded to hourly resolution; derived per-capita indicators are computed.

Step 7 — Remarks encoding: Categorical peak annotations are one-hot encoded (`peak_type_*`) and the original column dropped.

Step 8 — Missing value imputation: Bi-directional linear interpolation followed by forward/backward fills are executed on numeric features.

Step 9 — Target column creation: `target_demand_mw` is copied explicitly from `demand_mw`.

Step 10 — Lag and rolling computation: 8 lag and rolling variables are computed on the full dataset; initial warm-up rows (336 hours) are dropped.

Step 11 — Data partitioning: Dataset is chronologically split based on the experimental setup.

Step 12 — Feature scaling: Robust scaling is fitted strictly on training subsets. For neural networks, target standard scaling (`StandardScaler`) is applied.

The end-to-end processing pipeline is outlined in Figure 4.5.

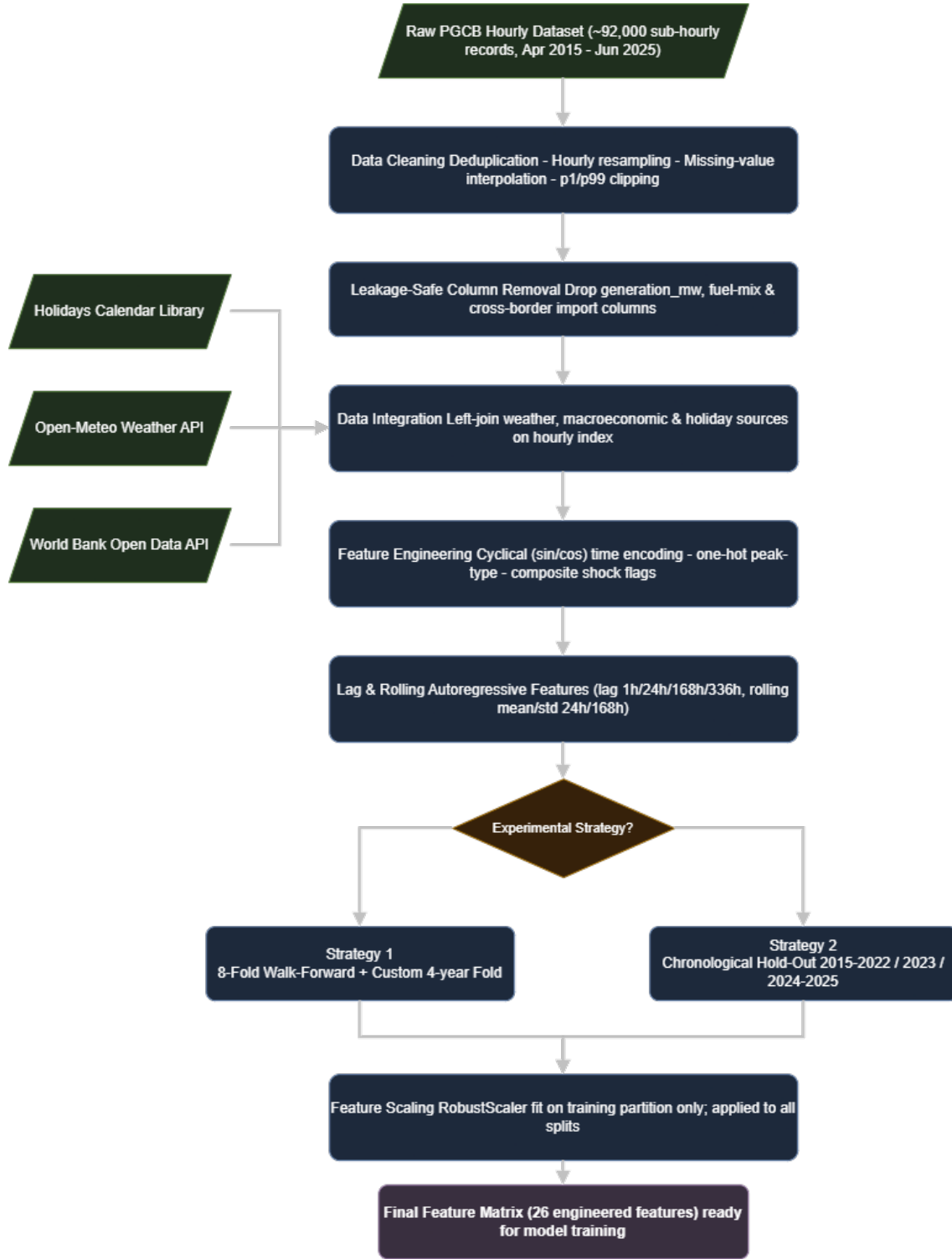


Figure 4.4: End-to-end preprocessing and modeling pipeline.

4.4.5 Evaluation Metrics

Model performance was evaluated using four standard metrics, summarized in Table 4.12. Mathematical definitions are specified below:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (4.4)$$

Table 4.12: Statistical Evaluation Metrics

Metric	Full Name	Purpose / Operational Focus
MAE	Mean Absolute Error	Measures average error magnitude with linear weighting.
RMSE	Root Mean Squared Error	Penalizes larger forecasting errors and outliers heavily.
MAPE	Mean Absolute Percentage Error	Scales error relative to actual demand magnitude.
R^2	Coefficient of Determination	Quantifies proportion of target variance explained.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (4.5)$$

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (4.6)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (4.7)$$

where y_i represents actual grid demand, $\hat{y}_i$ is forecasted demand, $\bar{y}$ is mean actual demand, and n is total evaluation hours.

4.4.6 Experimental Strategy 1: Chronological Hold-Out Split

The base experimental setup split data into Training (2015–2022; $\sim 70,000$ rows), Validation (2023; $\sim 8,760$ rows), and Testing (2024–2025; $\sim 12,000$ rows) sets.

All 7 traditional machine learning models and 7 deep learning architectures were evaluated under this setup:

- **Traditional ML Models:** Trained on scaled features (`X_train_scaled`, `y_train`). Early stopping (50 patience rounds) on the 2023 validation set was utilized for XGBoost and LightGBM.
- **Deep Learning Models:** Input tensors of shape (samples, 24, 26) were constructed with a 24-hour historical window. Trained using Adam optimizer (learning rate = 3×10^{-4} for LSTM, 1×10^{-4} for others) and MSE loss under gradient clipping (`clipnorm` = 1.0). Training ran up to 50 epochs with early stopping (patience = 10).

4.4.7 Hyperparameter Optimisation Framework

Hyperparameter optimization was executed on the LightGBM model using Optuna [47] over 100 trials with a Tree-structured Parzen Estimator (TPE) sampler, minimizing 2023 validation MAPE.

4.4.8 Experimental Strategy 2: Walk-Forward Temporal Validation

An 8-fold walk-forward validation strategy was executed across classical ML models. Additionally, a custom fold trained models on 2020–2023 ($\sim 35,064$ rows), validated on 2024 ($\sim 8,784$ rows), and evaluated performance on 2025 test data ($\sim 4,021$ rows), testing whether recent historical context better captures non-stationary structural shifts in grid demand post-COVID-19.

4.5 Optimisation Dataset Collection and Description

The Economic Load Dispatch (ELD) stage integrates three public data sources:

- **PGCB hourly generation and demand records (2025 subset) [11].** Supplies hourly demand (D_t) and generation by source (gas, liquid fuel, coal, hydro, solar, wind, imports) over January–June 2025 (3,994 usable hours).
- **BPDB Daily Electricity Generation Reports, 1 January–30 June 2025 (178 reports) [11].** Published daily by the Bangladesh Power Development Board. Extracted via PyMuPDF to capture:
 1. Evening peak shortfalls attributed to fuel limitations, low reservoir levels, or maintenance.
 2. Fuel-wise total generation cross-checks.
 3. Gas volumes supplied to power plants.
 4. Daily fuel costs by primary source.

Field extraction coverage exceeded 90%, extending historical daily report parsing approaches [48].

- **BPDB Annual Report 2024–25 [35].** Provides baseline marginal cost coefficients (b_i , in BDT/MWh) and cross-border purchase tariffs used for fiscal validation.

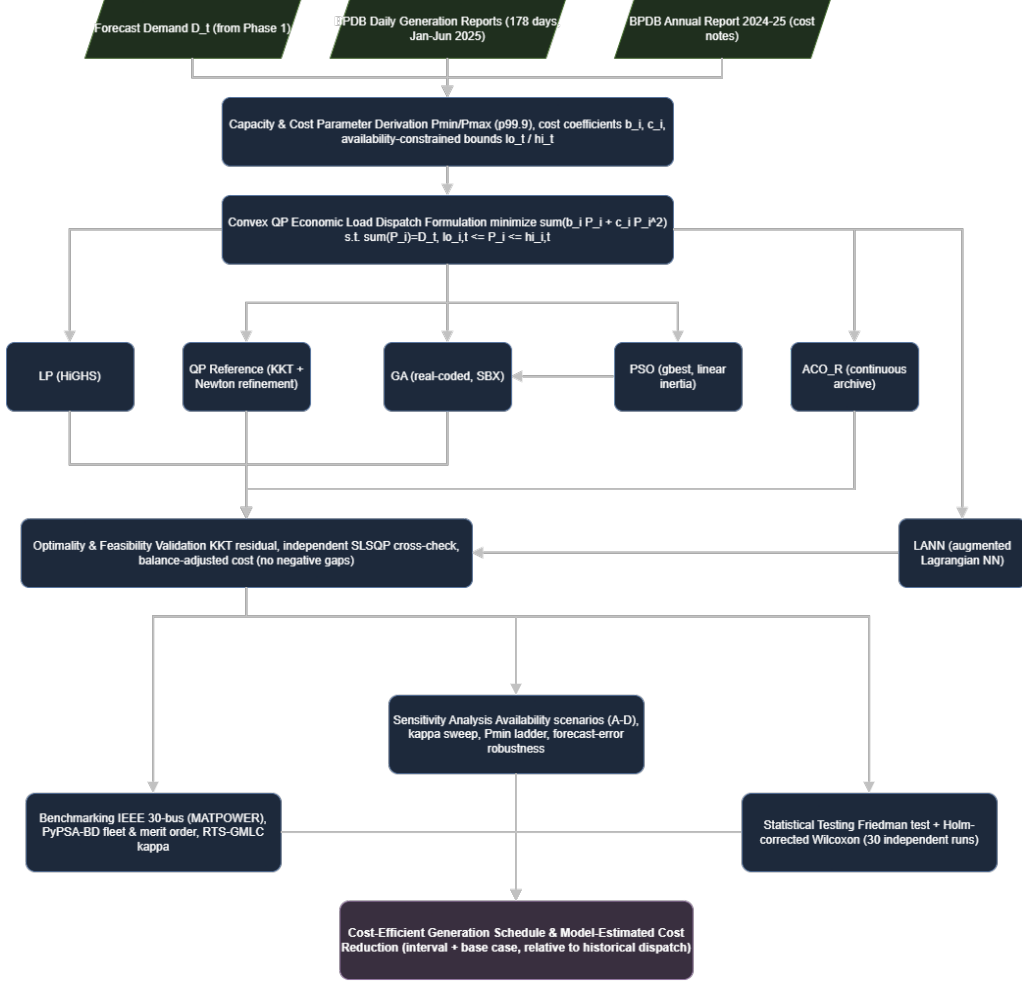


Figure 4.5: Optimization Pipeline

4.6 Economic Load Dispatch Problem Formulation

For each hour t , the ELD problem allocates target demand D_t across ten generation sources to minimize total fuel costs subject to capacity bounds:

$$\min \sum_i (b_i P_{i,t} + c_i P_{i,t}^2) \quad \text{subject to} \quad \sum_i P_{i,t} = D_t, \quad lo_{i,t} \leq P_{i,t} \leq hi_{i,t} \quad (4.8)$$

where b_i represents the linear cost parameter, c_i is a quadratic regularization term, and $[lo_{i,t}, hi_{i,t}]$ define hour-varying lower and upper operational capacity limits.

4.7 Optimisation Methods

Six optimization approaches were benchmarked on identical hourly dispatch instances:

- **Linear Programming (LP) reference:** Solved via HiGHS dual-revised simplex solver in SciPy [49] without quadratic terms.
- **Quadratic Programming (QP) reference:** Exact convex optimizer using bisection-Newton search on the Lagrange multiplier λ , cross-checked against SLSQP [18].

- **Genetic Algorithm (GA):** Real-coded GA with Simulated Binary Crossover (SBX) [50] and polynomial mutation.
- **Particle Swarm Optimization (PSO) [51]:** Global-best PSO with linearly decreasing inertia weight.
- **Continuous Ant Colony Optimization (ACO_R) [51]:** Continuous domain ACO algorithm using solution archives and Gaussian sampling.
- **Lagrangian Neural Network (LANN):** Deep feed-forward network trained with an augmented Lagrangian objective conditioned on $[D_t, \text{lo}_t, \text{hi}_t]$.

Stochastic methods (GA, PSO, ACO_R) were allocated a budget of 150,000 function evaluations and evaluated across 30 independent runs.

4.8 Optimisation Validation and Benchmarking Methodology

Validation followed a three-layer framework:

1. **Internal Optimality:** KKT residual verification and balance-adjusted cost normalization.
2. **Statistical Significance:** Friedman rank test followed by Holm-corrected pairwise Wilcoxon signed-rank tests across stochastic algorithms [52].
3. **External Benchmarking:** Benchmarked against the IEEE 30-bus test system (MATPOWER case30) [53], PyPSA-BD grid model [54], and NREL RTS-GMLC datasets [55] for parameter validation ($c_i, P_{\min}$) and sensitivity evaluation.

Chapter 5

Result Analysis

5.1 Performance Evaluation

Two families of evaluation metric are used in this chapter: one for the forecasting half of the study, already introduced in Section 4.4.5, and one for the dispatch-optimisation half, defined below.

5.1.1 Forecasting Evaluation Metrics

Forecasting accuracy is evaluated with the same four statistical metrics defined in Section 4.4.5 — Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and the coefficient of determination (R^2). MAPE is treated as the primary metric throughout this chapter because it is scale-independent and therefore comparable across folds and years in which the mean demand level itself changes.

5.1.2 Optimisation Evaluation Metrics

- **Optimality gap (%)**: For a method's dispatch cost C and the certified QP reference cost C_{QP} over the same sample of hours, the optimality gap is

$$\text{Gap\%} = 100 \times \frac{C - C_{QP}}{C_{QP}} \quad (5.1)$$

To prevent a mildly infeasible dispatch from appearing to beat the certified optimum, C is always the **balance-adjusted cost**: any residual power imbalance ($D_t - \sum_i P_{i,t}$) is priced at the QP's own marginal cost λ before the comparison is made,

$$C_{adj} = C_{raw} + \lambda \times \left(D_t - \sum_i P_{i,t} \right) \quad (5.2)$$

Under this definition the QP reference's own gap is exactly zero by construction, and every other method's gap is bounded below by that reference.

- **Balance error (MW)**: The mean absolute power-balance violation $|D_t - \sum_i P_{i,t}|$ for a given method.
- **Constraint violations**: The count and magnitude of any dispatch that falls outside its source's lower/upper bound.

- **Model-estimated cost reduction (%)**:

$$\text{Reduction\%} = 100 \times \frac{\text{Cost}_{\text{actual}} - \text{Cost}_{\text{optimized}}}{\text{Cost}_{\text{actual}}} \quad (5.3)$$

computed over the historical validation window, and always reported as a base case together with a sensitivity range rather than as a single figure (Section 5.2.2).

- **Unit cost (BDT/kWh)**: Total dispatch cost divided by total energy served, used to express the cost reduction in tariff-comparable terms.
- **Statistical significance**: A Friedman test across the stochastic methods (GA, PSO, ACO), each run 30 times, followed by Holm-corrected pairwise Wilcoxon signed-rank tests [52] (Section 5.4).

5.2 Analysis of Design Solutions

5.2.1 Forecasting Results

Experimental Strategy 1: Chronological Hold-Out Split Results

The full results for Experimental Strategy 1 are given in Table 5.1.

Table 5.1: Experimental Strategy 1 — Full Model Comparison (Chronological Split)

Model	Train MAPE (%)	Val MAPE (%)	Test MAPE (%)	Test MAE (MW)	Test RMSE (MW)	Test R^2	Overfit Label
Linear Regression	5.70	5.35	5.10	411.13	1601.50	0.6506	LOW
Ridge Regression	5.70	5.46	5.30	426.28	1606.23	0.6485	LOW
Lasso Regression	5.70	5.45	5.28	424.33	1605.62	0.6488	LOW
Random Forest	4.47	4.41	4.70	393.35	1453.89	0.7120	LOW
Gradient Boosting	3.99	4.49	5.01	444.09	1494.40	0.6957	LOW
XGBoost	3.82	4.15	4.73	409.08	1472.79	0.7045	LOW
LightGBM	3.78	4.01	4.47	378.21	1458.12	0.7103	LOW
LSTM	4.52	5.26	5.31	471.29	1502.16	0.6923	LOW
BiLSTM	9.99	10.98	11.38	1168.37	2114.61	0.3902	LOW
GRU	10.40	10.56	11.09	1134.95	2019.26	0.4440	LOW
BiGRU	8.83	9.22	9.64	940.08	1920.53	0.4970	LOW
TCN	9.99	9.80	10.24	1014.10	1867.48	0.5244	LOW
CNN-BiLSTM	10.10	10.11	10.84	1103.92	2009.37	0.4494	LOW
Attention-LSTM	12.76	12.58	13.39	1425.36	2313.67	0.2700	LOW

End of table

Experimental Strategy 1 had the following key observations. First of all, all traditional ML models have low overfitting (MAPE gap < 1.5%), which indicates that the model is well regularized by the regularisation procedures and the large size of the training set. Secondly, LightGBM performs the best MAPE on the test set among all the architectures, with a MAPE of 4.47% ($R^2 = 0.7103$), followed by Random Forest (4.70%) and XGBoost (4.73%). Third, when comparing the performance of different deep learning models with

that of the traditional ML models, the conventional LSTM models come close to the performance of the traditional ML models ($\text{MAPE} = 5.31\%$, $R^2 = 0.6923$), thereby indicating that the current deep learning architectures and the look-back window of 24 hours do not fully leverage the temporal structure in this dataset. The results of these analyses all support the choice of LightGBM as the main model to optimize its hyperparameters. Figure 5.1 shows the actual versus predicted electricity demand produced by the tuned LightGBM model on the 2024–2025 test period, illustrating the model’s ability to reproduce diurnal peak-trough cycles and weekend demand suppression.

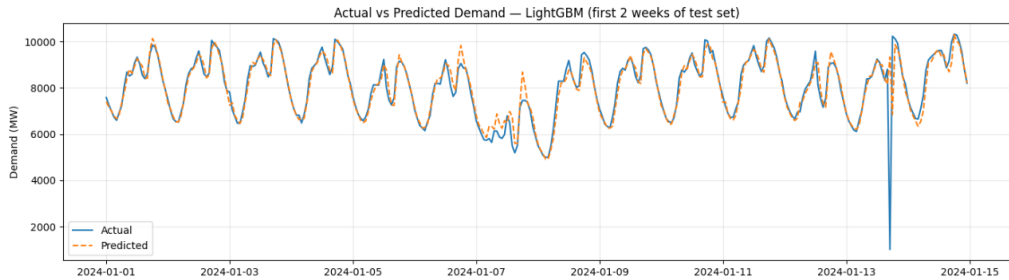


Figure 5.1: Actual versus predicted electricity demand (MW) produced by LightGBM on the 2024–2025 test period. The weekly zoom inset illustrates the model’s ability to reproduce diurnal peak–trough cycles and weekend demand suppression.

Hyperparameter Optimisation Framework Results

After performing the baseline comparison, the hyperparameter optimisation was carried out on the LightGBM model using the Optuna framework [47] with 100 trials and the default Tree-structured Parzen Estimator (TPE) sampler. The goal function minimizes the validation MAPE for the hold-out year 2023. The early stopping (50 rounds) is used to avoid over-fitting during the search within each trial.

The hyperparameter search space and the optimal settings found after 100 trials is shown in Table 5.2.

Table 5.2: LightGBM Optuna Hyperparameter Search Space and Optimal Configuration

Hyperparameter	Search Range	Optimal Value	Description
n_estimators	100 – 2,000	1,900	Number of boosting rounds (trees)
learning_rate	0.005 – 0.300	0.0277	Step size shrinkage applied per tree
max_depth	3 – 12	9	Maximum depth of individual trees
num_leaves	8 – 256	20	Maximum number of leaves per tree
min_child_samples	5 – 100	57	Minimum samples required in a leaf node
subsample	0.4 – 1.0	0.657	Row sub-sampling ratio per tree

Continued on next page

Table 5.2: LightGBM Optuna Hyperparameter Search Space and Optimal Configuration (continued)

Hyperparameter	Search Range	Optimal Value	Description
colsample_bytree	0.4 – 1.0	0.999	Feature sub-sampling ratio per tree
reg_alpha	0 – 5	0.218	L1 regularisation on leaf weights
reg_lambda	0 – 5	0.000182	L2 regularisation on leaf weights

End of table

An optimisation was carried out to the best validation MAPE of 3.88% at Trial 91. When tested on the 2024–2025 test set, the tuned model has a test MAPE of 4.30% and an R^2 of 0.7129, which is a gain of 0.17 percentage point when compared with the baseline LightGBM model. This mild improvement indicates that the default hyperparameters are already relatively good for this dataset, and deeper hyperparameter search may not be the best approach to improve performance, at least for this dataset, unless improvements in the feature set or additional training data are found.

Table 5.3 compares the validation and test performance of the baseline LightGBM model against the Optuna-tuned configuration.

Table 5.3: LightGBM Baseline vs. Optuna-Tuned Performance

Configuration	Best Val MAPE (%)	Test MAPE (%)	Test R^2
LightGBM (baseline)	4.01	4.47	0.7103
LightGBM (Optuna-tuned)	3.88	4.30	0.7129
Improvement	−0.13	−0.17	+0.0026

End of table

Experimental Strategy 2: Walk-Forward Temporal Validation Results

Table 5.4 shows the per-fold results of all the conventional ML models. Fold 1 is kept for completeness but not used for the aggregate score as it is found that the distribution has changed over time, as mentioned in Section 4.4.2.

Table 5.4: Walk-Forward Validation — Per-Fold Test MAPE (%) for All Models

Fold (Test Year)	Lin. Reg.	Ridge	Lasso	Rand. Forest	Grad. Boost.	XGBoost	LightGBM
Fold 1 — 2018	16.62	16.61	16.62	17.26	17.08	17.85	17.19
Fold 2 — 2019	5.27	5.27	5.27	6.24	6.84	7.66	7.08
Fold 3 — 2020	6.03	6.03	6.03	3.82	4.23	3.90	3.91
Fold 4 — 2021	3.09	3.09	3.09	2.55	2.52	2.43	2.27
Fold 5 — 2022	3.40	3.40	3.40	3.87	3.78	3.74	3.64

Continued on next page

Table 5.4: Walk-Forward Validation — Per-Fold Test MAPE (%) for All Models (continued)

Fold (Test Year)	Lin. Reg.	Ridge	Lasso	Rand. Forest	Grad. Boost.	XGBoost	LightGBM
Fold 6 — 2023	4.94	4.94	4.94	4.92	5.14	4.81	4.69
Fold 7 — 2024	5.44	5.44	5.44	5.15	5.47	5.19	5.11
Fold 8 — 2025	4.53	4.53	4.53	2.57	2.93	2.82	2.66

End of table

Note: Fold 1 was removed from the aggregate score due to distributional shift (see Section 4.4.2).

The overall performance of the seven reliable folds (Folds 2-8) is presented in Table 5.5. The standard deviation of test MAPE over the folds gives a measure of temporal stability – smaller the standard deviation, more stable the generalisation across different demand regimes.

Table 5.5: Walk-Forward Summary — Folds 2–8 Average Performance

Model	Avg. Val MAPE (%)	Avg. Test MAPE (%)	Std. Test MAPE (%)	Avg. Test R^2	Avg. Test MAE (MW)
Random Forest	5.22	4.16	± 1.37	0.8320	368.6
LightGBM	5.20	4.19	± 1.63	0.8235	374.4
XGBoost	5.25	4.36	± 1.75	0.7965	394.3
Gradient Boosting	5.24	4.42	± 1.51	0.7597	402.7
Lasso	6.40	4.67	± 1.08	0.8085	384.5
Linear Regression	6.40	4.67	± 1.08	0.8085	384.5
Ridge	6.40	4.67	± 1.08	0.8085	384.5

End of table

Random Forest has the lowest average test MAPE in all Folds 2-8 (4.16%, $R^2 = 0.8320$) and the fold-to-fold std. dev. is $\pm 1.37\%$, showing good uniformity in generalisation across various demand years. LightGBM is similar (4.19%, $R^2 = 0.8235$) with slightly higher variability (1.63%). The gradient boosting models and the XGBoost models are the two top contenders for ensemble leadership, and the linear models provide the most consistent and less precise forecasts (MAPE $\approx 4.67\%$, std $\approx \pm 1.08\%$). The MAPE rank of Gradient Boosting (0.7597) is generally lower than of its R^2 value indicates, meaning that although the mean percentage error is acceptable, the model does not perform as well when explaining variations.

Figure 5.2 illustrates the test MAPE for each walk-forward fold (Folds 2–8) across all seven traditional ML models, where the grey shaded band represents the min-max range of the four tree-based ensemble models per fold.

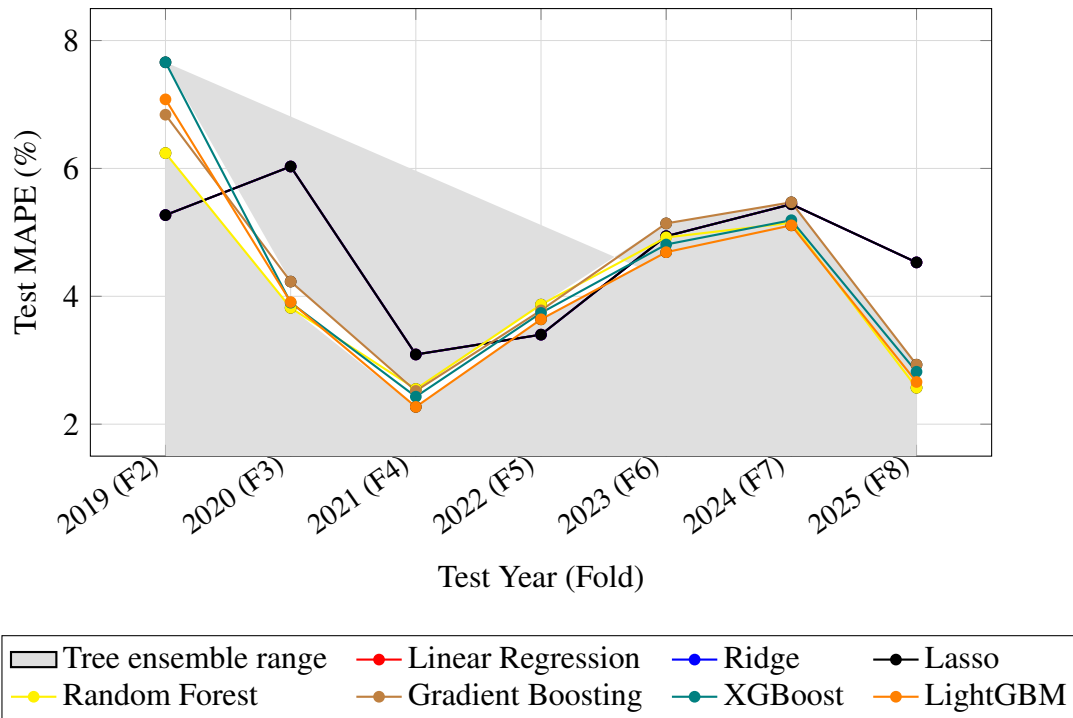


Figure 5.2: Test MAPE for each walk-forward fold (Folds 2–8) across all seven traditional ML models. The horizontal axis represents the test year (2019–2025). Gray shaded band indicates the range (min–max) of the four tree-based ensemble models (Random Forest, Gradient Boosting, XGBoost, LightGBM) per fold.

Custom Fold Analysis: Recent-Data Training Hypothesis

A supplemental custom fold was created to see if this focus on the most recent four years of historical data (versus the two-year windows in the standard folds) would yield better forecasting accuracy on the next four years of test data. This hypothesis is driven by the fact that the time-series of Bangladesh demand is not stationary, as structural changes in grid capacity are likely to have included sustained additions in the grid, economic recovery from COVID-19, and changing consumption patterns, all of which are expected to have introduced structural variations in the grid that pre-2020 observations are less likely to capture.

The custom fold trains all models on 2020–2023 (~35,064 rows), validates on 2024 (~8,784 rows), and evaluates on the 2025 test period (~4,021 rows). Table 5.6 shows the results of this.

Table 5.6: Custom Fold Results — Training 2020–2023, Validation 2024, Test 2025

Model	Val MAPE (%) [2024]	Test MAPE (%) [2025]	Test R^2	Test MAE (MW)	Test RMSE (MW)
Random Forest	4.76	2.65	0.9622	305.5	414.4
LightGBM	5.07	2.68	0.9613	311.3	419.5
XGBoost	5.29	3.38	0.8361	412.3	863.1
Gradient Boosting	5.27	3.35	0.7822	409.7	994.9

Continued on next page

Table 5.6: Custom Fold Results — Training 2020–2023, Validation 2024, Test 2025 (continued)

Model	Val MAPE (%) [2024]	Test MAPE (%) [2025]	Test R^2	Test MAE (MW)	Test RMSE (MW)
Ridge	6.35	3.84	0.9221	430.2	595.2
Lasso	6.35	3.84	0.9220	430.3	595.2
Linear Regression	6.35	3.84	0.9220	430.3	595.2

End of table

The results support the recent data training hypothesis. The results are obtained on the 2025 test set, with test MAPE of 2.65% ($R^2 = 0.9622$) for Random Forest and 2.68% ($R^2 = 0.9613$) for LightGBM, which is nearly 1.5 percentage points better than the respective folds 2–8 averages. The models trained over a longer period of four years have substantially higher R^2 values (> 0.96), demonstrating that they are better generalised to the conditions of 2025, given that they include the recovery surge in 2021 and the economic stress period in 2022. The results have a practical significance, because a rolling four-year training window may be preferred to two-year sliding folds in operation, which will be formally confirmed in the next phase of research.

Figure 5.3 compares the test MAPE on the 2025 test period between the standard Folds 2–8 average and the custom four-year training fold for Random Forest and LightGBM, demonstrating the benefit of incorporating a longer, more recent training window.

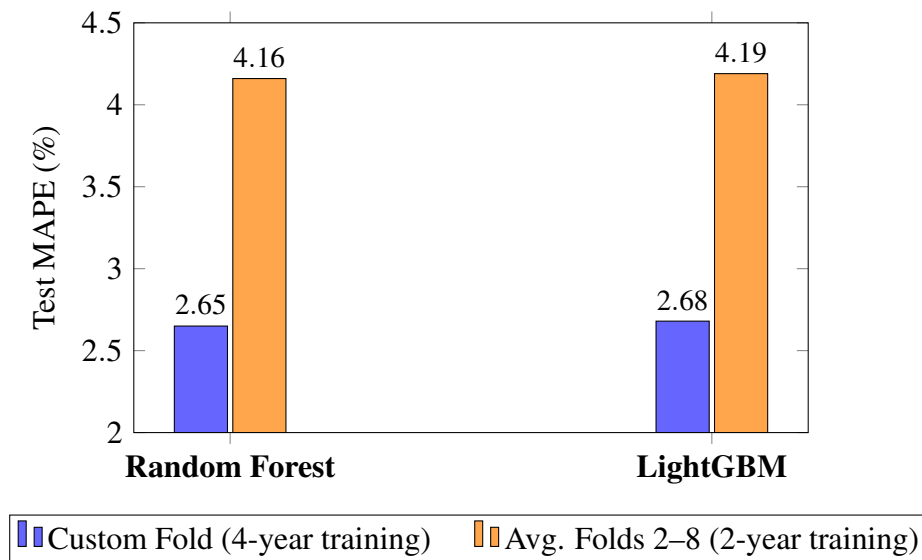


Figure 5.3: Comparison of test MAPE on the 2025 test period between the average of Folds 2–8 (2-year training windows) and the Custom Fold (4-year training window) for Random Forest and LightGBM. The substantial reduction in MAPE (from $\approx 4.2\%$ to $\approx 2.7\%$) demonstrates the benefit of incorporating a longer, more recent four-year training window.

5.2.2 Optimisation / Economic Dispatch Results

Main algorithm comparison. Table 5.7 compares all six dispatch methods on a stratified 24-hour sample, with GA, PSO and ACO averaged over 30 independent runs. The QP reference is, by construction, the zero-gap certified optimum. ACO and PSO track it most closely (mean gap 0.0003% and 0.0033% respectively), the LANN is close behind at 0.0094% despite solving the problem with a single forward pass rather than an iterative search, and GA is the weakest of the six at 0.48%. All balance errors are below 14 MW against a mean hourly demand above 10,000 MW, and no method records a constraint violation.

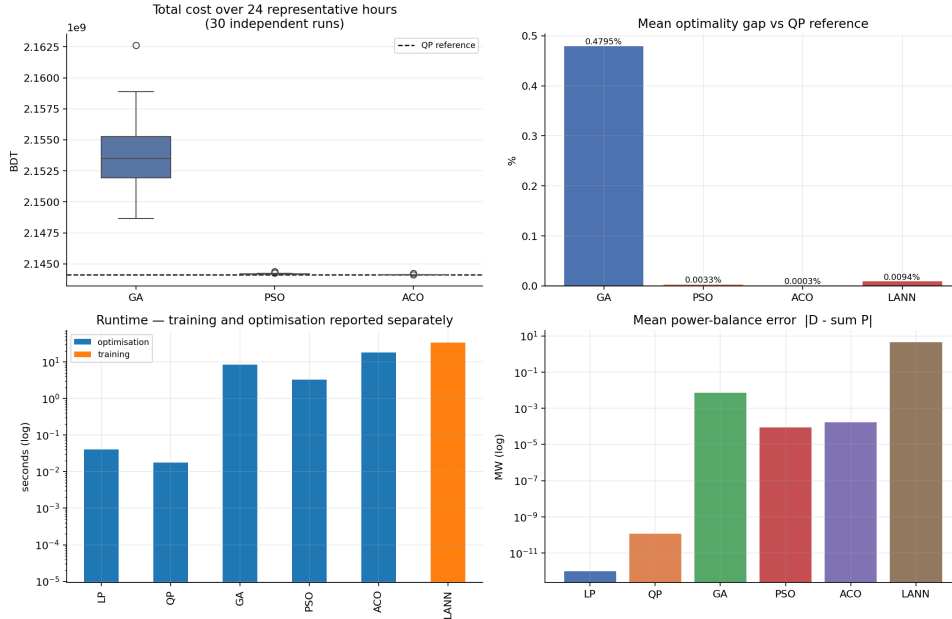


Figure 5.4: Algorithm comparison over the 24-hour stratified sample — total cost distribution (GA/PSO/ACO, 30 runs), mean optimality gap vs. the QP reference, runtime, and mean power-balance error

Table 5.7: Main dispatch method comparison (24-hour stratified sample, 30 runs for GA/PSO/ACO)

Method	Mean gap vs. QP (%)	Mean balance error (MW)	Notes
LP	0.038	≈ 0	Linear objective; not directly comparable on the quadratic cost
QP (reference)	0.000	≈ 0	Certified KKT optimum
GA	0.480	0.007	30 runs; weakest of the six methods
PSO	0.0033	0.00009	30 runs
ACO	0.0003	0.00017	30 runs; closest to QP

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Table 5.7: Main dispatch method comparison (continued)

Method	Mean gap vs. QP (%)	Mean balance error (MW)	Notes
LANN	0.0094	4.59 (mean), 13.7 (max)	Single forward pass per hour; $\approx 500\times$ faster inference than re-solving the QP

End of table

Estimated cost reduction and its sensitivity to fuel availability. The single most important design decision in the optimisation layer is how much gas and liquid-fuel capacity is assumed available, because BPDB’s own daily reports show that a meaningful share of nominal capacity is routinely unavailable for lack of fuel. Table 5.8 reports the resulting availability ladder: an unconstrained assumption (scenario A) yields an unrealistically large 34.6% reduction, most of which simply reflects gas that historical BPDB reports say was never available; applying BPDB’s own reported daily limitation (scenario B2) gives the base-case result of 21.27%; and holding gas at exactly its historical hourly output (scenario D, the most conservative possible assumption) still yields 11.39% — the irreducible allocative saving that survives even under the most conservative fuel-availability assumption.

Table 5.8: Model-estimated cost reduction under different fuel-availability assumptions

Scenario	Description	Optimised cost (bn BDT)	Reduction (%)
A	Unconstrained (Section 4.5 bounds only)	287.69	34.63
B1	BPDB reported variation, $C_{ref} - (L_d - L_{min})$	338.79 (prorata) / 364.21 (gas-only)	23.02 / 17.24
B2 (base case)	BPDB reported level, $C_{ref} - L_d$	346.46 (prorata) / 373.50 (gas-only)	21.27 / 15.13
C	Observed gas+LF + 10% headroom	385.08	12.50
D	Gas+LF fixed at actual (fully binding)	389.95	11.39

End of table

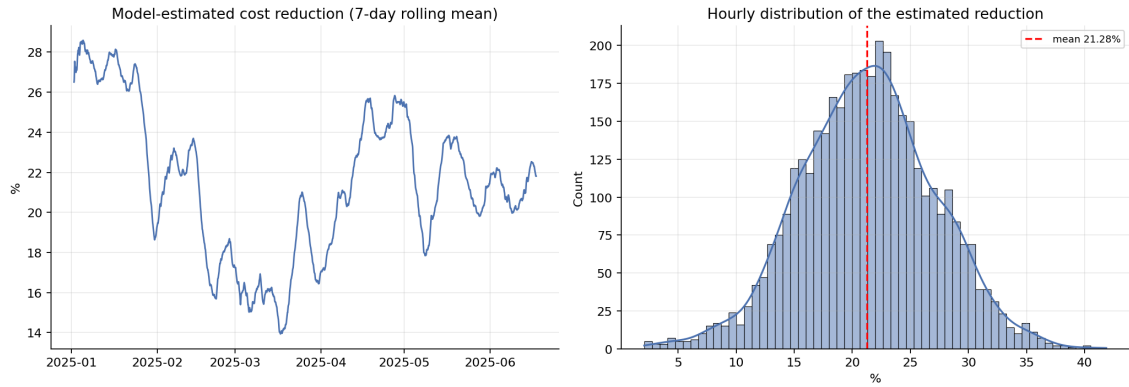


Figure 5.5: *Model-estimated cost reduction over the study window — 7-day rolling mean over time (left) and hourly distribution with mean 21.28% (right), under the base-case (B2) availability assumption*

Historical validation, generation mix and cost trend. Against actual 2025 dispatch, the base-case optimisation reduces total cost from 440.08 to 346.45 billion BDT over the 3,994-hour study window (approximately 46.1 TWh), an average unit-cost reduction from 9.55 to 7.52 BDT/kWh. The optimiser reaches this by shifting generation share toward the least expensive domestic sources: gas rises from 45.1% to 55.2% of energy and coal from 28.2% to 29.1%, while the costlier liquid-fuel/HFO share falls from 9.1% to 0.9% and the more expensive import links (Adani) fall from 7.8% to 3.0% of energy, with cheaper imports (Nepal, Tripura) increasing modestly.

Table 5.9: Generation mix, actual vs. optimised (energy share, %)

Source	Actual share (%)	Optimised share (%)	Change (pp)
Gas	45.09	55.22	+10.13
Liquid Fuel / HFO	9.09	0.90	-8.19
Coal	28.21	29.14	+0.93
Hydro	0.45	0.46	+0.01
Solar	1.30	1.30	0.00
Wind	0.08	0.08	0.00
Bheramara HVDC (import)	7.60	8.62	+1.02
Tripura (import)	0.40	0.97	+0.57
Adani (import)	7.77	2.96	-4.81
Nepal (import)	0.01	0.34	+0.33

End of table

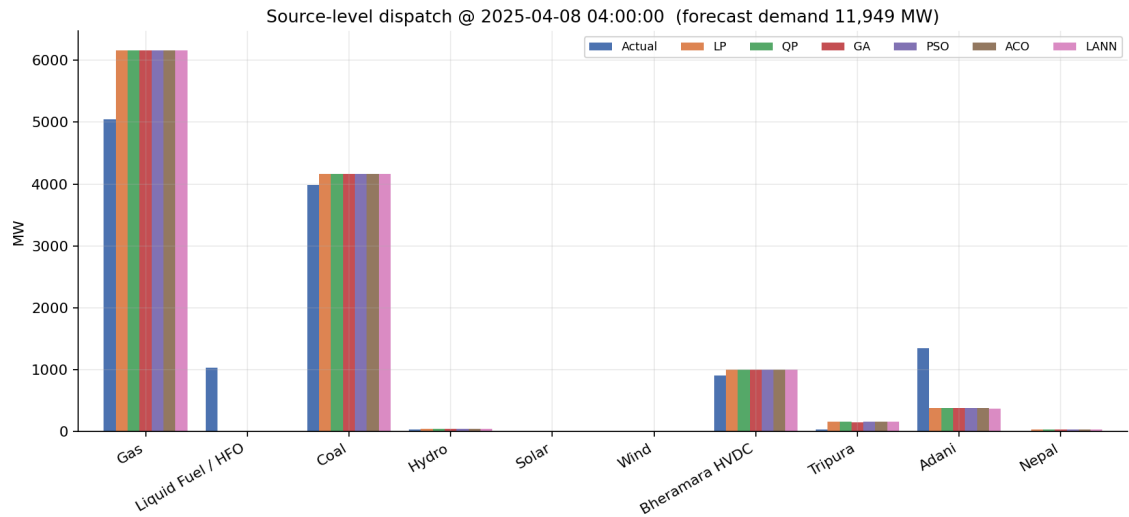


Figure 5.6: Hourly dispatch by source, actual versus optimised

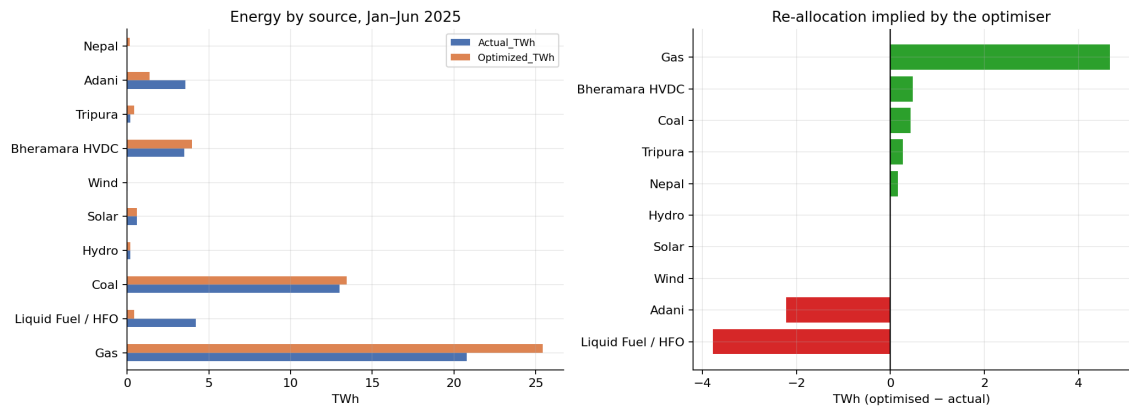


Figure 5.7: Generation mix by source, actual vs. optimised energy share

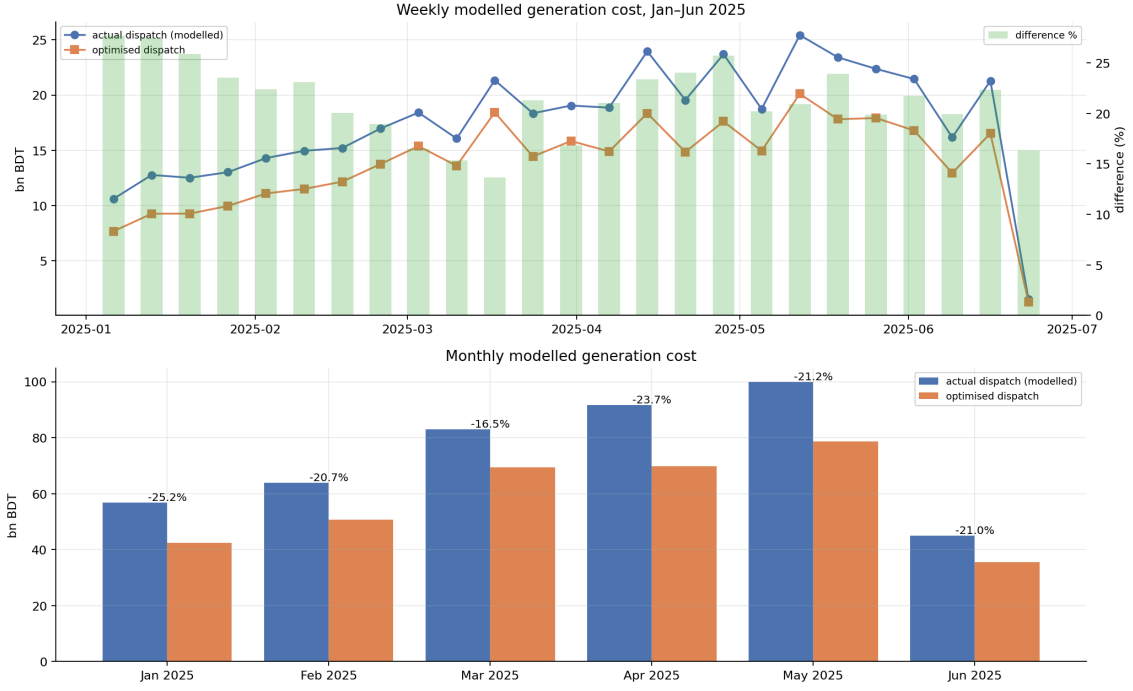


Figure 5.8: Weekly and monthly actual vs. optimised dispatch cost trend across the study window

The monthly trend (Table 5.10) shows the reduction is largest early in the study window (25.2% in January) and gradually narrows toward June (21.0%) as demand rises seasonally and the cheapest sources approach their availability-constrained bounds more often.

Table 5.10: Monthly cost reduction trend

Month	Actual cost (bn BDT)	Optimised cost (bn BDT)	Reduction (%)
Jan 2025	56.74	42.46	25.16
Feb 2025	63.83	50.62	20.69
Mar 2025	83.03	69.30	16.53
Apr 2025	91.61	69.88	23.73
May 2025	99.94	78.72	21.23
Jun 2025 (partial)	44.93	35.47	21.04

End of table

5.2.3 LANN Training Behaviour and Convergence

Figure 5.9 shows the GA/PSO/ACO convergence curves and Figure 5.10 the LANN's training history. The LANN's augmented-Lagrangian loss decreases smoothly once the network is conditioned on the full problem instance $[D, lo, hi]$ rather than on demand alone, with the penalty weight (ρ) escalating only while the power-balance constraint remains materially violated and then holding once the model is within its feasibility tolerance.

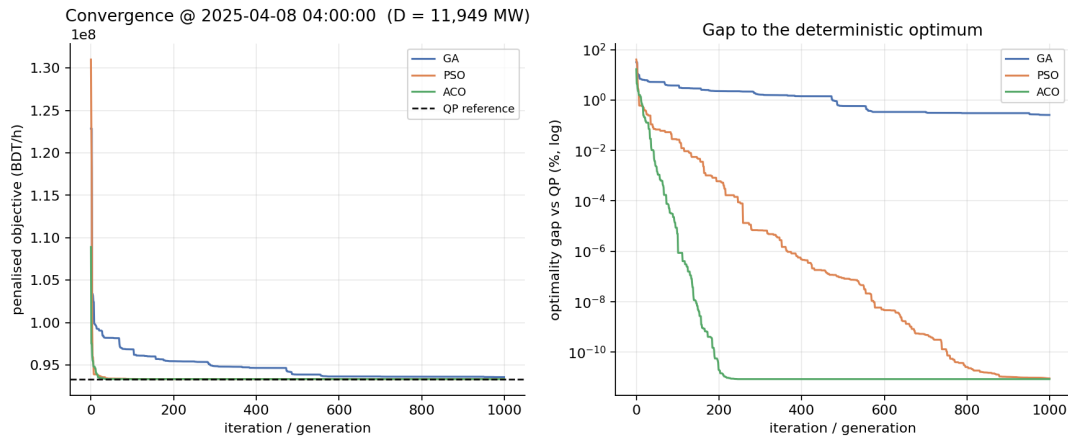


Figure 5.9: *GA/PSO/ACO convergence over the optimisation budget*

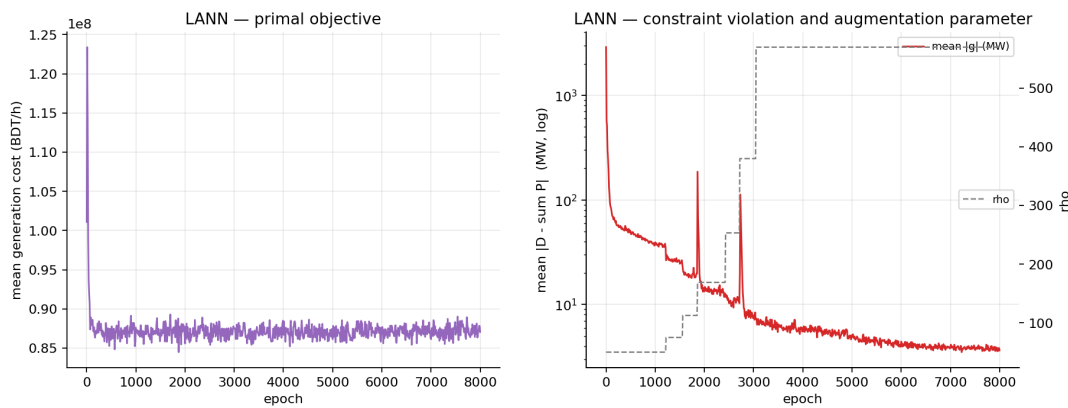


Figure 5.10: *LANN training history — cost, mean constraint violation, penalty weight (ρ) and mean dual variable (λ) versus training epoch*

5.3 Final Design Adjustments

Several design corrections were made after an initial implementation produced results that did not survive internal scrutiny. They are recorded here as evidence of an iterative, verification-driven design process.

Table 5.11: Design iterations and their effect on the result

Initial design	Problem found	Adjustment made	Effect on result
Unconstrained source bounds	Headline reduction of 34.6% relied on gas capacity BPDB's own reports say was unavailable	BPDB daily-reported fuel-availability limitation applied as an hour-varying upper bound	Base case revised to 21.27%, reported as an 11.39–34.63% range rather than a single number

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Table 5.11: Design iterations and their effect on the result (continued)

Initial design	Problem found	Adjustment made	Effect on result
Capacity parameters from a single later snapshot report	Snapshot postdated the Jan–Jun 2025 study window by about a year	Capacity re-derived from the 178 in-window daily reports themselves	Removes the temporal mismatch; the interconnection capacities were confirmed unchanged, so headline figures were unaffected but are now correctly sourced
Raw generation cost used for every method’s optimality gap	A mildly infeasible dispatch could show a lower raw cost than the certified optimum, producing a negative, meaningless gap	Balance-adjusted cost introduced — any residual imbalance priced at the QP’s own marginal cost before comparison	All reported gaps are now non-negative and the QP reference’s own gap is pinned to exactly zero
LANN conditioned on demand only (plus solar/wind)	Once hour-varying availability bounds were introduced, the network could not tell which problem instance it was solving; gap rose to 9.4% with a 597 MW mean imbalance	Network re-conditioned on the full instance $[D, lo, hi]$; penalty schedule hardened (lower cap, feasibility-gated escalation, best-checkpoint retention)	Gap fell to 0.0094%, mean balance error to about 5 MW, and the learned dual variable now correlates with the true marginal cost
Ranking sensitivity sweep priced imbalance at the maximum marginal cost across sources	Over-prices infeasibility and can itself produce a negative gap	Re-priced using the QP’s own certified shadow price	Sensitivity-sweep gaps are consistent with the main comparison table

End of table

5.4 Statistical Analysis

Optimisation methods. The 30 independent runs of GA, PSO and ACO on the 24-hour stratified sample were compared with a Friedman test across the three algorithms’ total costs, followed by Holm-corrected pairwise Wilcoxon signed-rank tests [52]. The Friedman test rejects the null hypothesis of equal performance overwhelmingly ($\chi^2 = 58.07$, $p < 10^{-12}$, $df = 2$, $n = 30$), and every pairwise comparison remains significant after Holm correction (GA vs. PSO: $p < 10^{-8}$; GA vs. ACO: $p < 10^{-8}$; PSO vs. ACO: $p < 10^{-5}$), so the ranking $ACO < PSO \ll GA$ (lower cost is better) is not attributable to

run-to-run noise.

Table 5.12: Friedman and Holm-corrected Wilcoxon test results (30 runs, 24-hour sample)

Test	Statistic	p -value	Conclusion
Friedman (GA, PSO, ACO)	$\chi^2 = 58.07$ ($df = 2$)	$< 10^{-12}$	Performance differs significantly across the three algorithms
Wilcoxon, GA vs. PSO (Holm-corrected)	$W = 0$	$< 10^{-8}$	Significant
Wilcoxon, GA vs. ACO (Holm-corrected)	$W = 0$	$< 10^{-8}$	Significant
Wilcoxon, PSO vs. ACO (Holm-corrected)	$W = 24$	$< 10^{-5}$	Significant

End of table

Forecasting models. Table 5.5 already reports the fold-to-fold standard deviation of walk-forward test MAPE for every model: Random Forest is both the most accurate (4.16%) and among the most stable (std. dev. $\pm 1.37\%$) of the tree-based ensembles, while the linear models are the least accurate but most stable ($\pm 1.08\%$), a variance/accuracy trade-off worth noting rather than a contradiction.

5.5 Comparisons and Relationships

5.5.1 Benchmarking against independent systems

IEEE 30-bus (MATPOWER case30) [53]. The same six dispatch methods were re-run with the genuine literature source generator cost coefficients instead of the BPDB derived cost coefficients used in this study on IEEE 30 bus problem. The ranking result was perfectly aligned with the expectations. LP is the most deviated one with 14.43% far off from the quadratic objective optimum. On the other hand, GA (0.042%), PSO (0.0035%), ACO (0.001%), and the LANN (0.0005%), all was under 0.05% which confirms that this study's own cost-coefficient derivation didn't affect on the result of the comparison's conclusion.

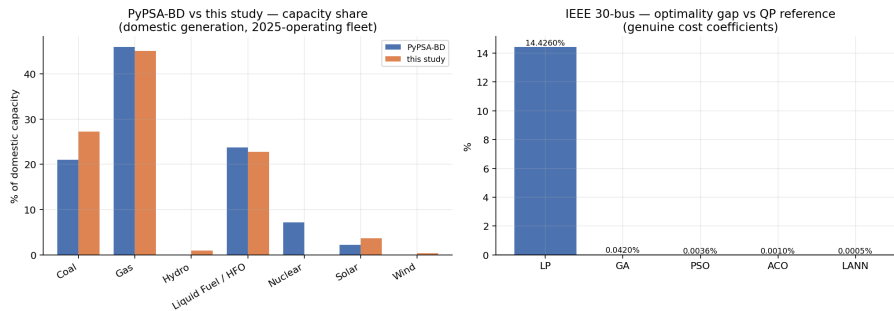


Figure 5.11: *External benchmarking - PyPSA-BD vs. this study's domestic capacity share (left) and IEEE 30-bus optimality gaps with genuine cost coefficients (right)*

PyPSA-BD [54]. A localised grid model in Bangladesh which was independently published. The domestic generation-capacity shares are similar to the shares of this study derived from the BPDB (gas 45.9% vs. 45.1%, coal 21.0% vs. 27.3%, liquid fuel 23.7% vs. 22.7%). The merit order implied by its cost assumptions is highly correlated (Spearman rho = 0.97, p = 0.005, n = 5 technologies) with the merit order of this study based on BDT though not being EUR dominated. This is used as a rank-order cross-check and is not an independent cost source, as absolute currencies and cost concepts are not comparable.

RTS-GMLC [55]. This study finds a fleet-level BPDB-derived estimate of the quadratic cost coefficient, median kappa $\tilde{0}$, with an IQR of 0.20-0.58, lower than the NREL synthetic test system's estimate, which provided an estimate of the quadratic cost coefficient based on literature values for each thermal unit, kappa $\tilde{0}$.38 (IQR 0.20-0.58) across 64 thermal units. The value of kappa that has been adopted (0.1) is located between the range [0.01, 0.40] which is very moderate (between 18.5% and 22.4%) under the base case availability scenario, therefore the outcome is not sensitive to this modelling choice.

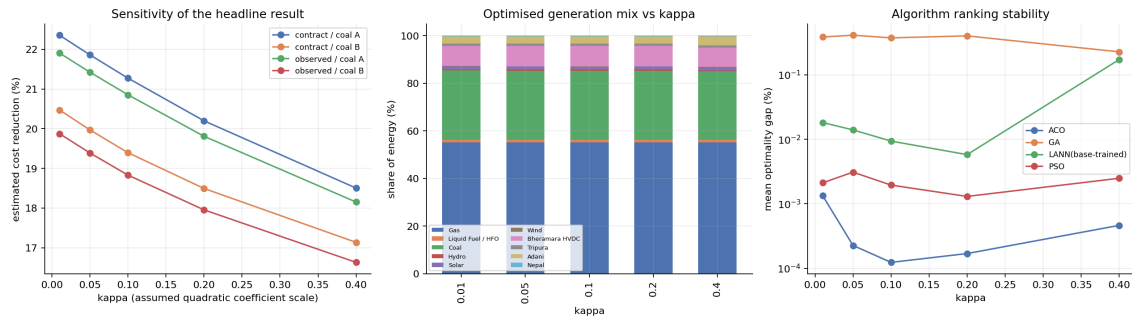


Figure 5.12: Sensitivity of the estimated cost reduction to the quadratic cost coefficient (kappa), import cost basis and coal scenario

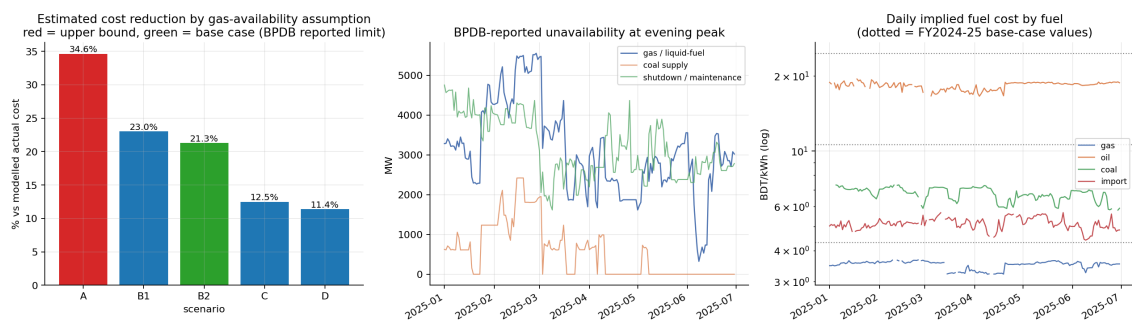


Figure 5.13: Availability scenarios and BPDB-reported daily fuel cost series used to derive the model's cost and capacity parameters

Data cross-check. This PGCB hourly series, used both for forecasting and optimization, is independently validated against BPDB's separately calculated daily totals of energy at $r = 0.994 - 0.998$, supplied by gas, coal, oil, solar and imports (178 days overlap) which is the main line of defence against the prospect that the PGCB dataset is unreliable.

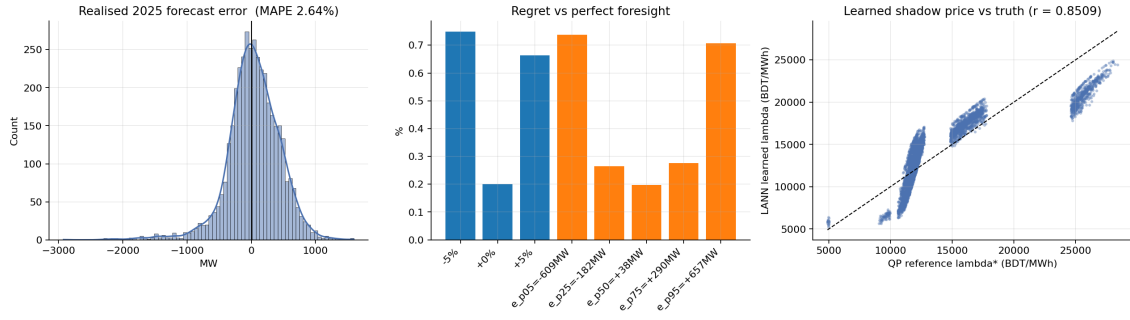


Figure 5.14: *Forecast-error robustness (regret vs. forecast bias) and the LANN's learned dual variable tracking the true system marginal cost*

5.5.2 Relationship between forecast accuracy and dispatch cost

The report of the forecast-error robustness dispatch (in Table 5.13) is planned using a forecast of demand with a systematic or an empirically sampled error, and then re-dispatched against the actual demand. The total cost incurred is compared with the cost that can be achieved if the dispatch is perfect.

Table 5.13: Forecast-error robustness (regret relative to perfect-foresight dispatch)

Forecast error scenario	Total cost (bn BDT)	Regret vs. perfect foresight (%)
-5% (systematic under-forecast)	352.28	0.75
+0% (unbiased)	350.36	0.20
+5% (systematic over-forecast)	351.98	0.66
Empirical p05 (-609 MW)	352.24	0.74
Empirical p50 (+38 MW)	350.35	0.20
Empirical p95 (+657 MW)	352.13	0.71

Regret stays within the range of 0.75% of total cost for all the range of forecasting errors tested, including forecasting errors that are several times larger than the forecasting model's observed accuracy. The relationship is the sharpest of all the results and is actually discussed in Section 5.6, both at the level of demand volatility at this system and at the level of accuracy achieved by the forecasting model. It is an operationally useful finding, concerning the place where forecasting efforts are not that impactful and where they are likely to be valuable, rather than as a weakness of the integration.

5.6 Discussions

Combined together, the results confirm three primary hypotheses. Firstly, the dispatch optimum used in this study as a reference is not just claimed but verified: the KKT stationarity residual for the KKT system of the dispatch optimum in the formulation is zero to machine precision and reproduced by a general purpose SLSQP solver. It makes it easy to ensure that all comparisons in this chapter are made against a verified (not simply a reported) optimum. Secondly, the headline figure for cost reduction is not sensitive to any modelling assumption that could be identified rather than measurement (quadratic cost coefficient, minimum-generation rule, forecast error). These can only shift the figure by a few percentage points and to the extent that the headline figure is sensitive to

any one modelling choice (how much gas capacity is assumed available). That one modelling choice was a data-supported range, not a single number, and not an assumption. Thirdly, among the six dispatch methods compared, GA is outperformed by the classical, gradient-free metaheuristics (PSO and ACO) and QP/LP methods on optimality. It even outperformed the learned LANN. But LANN's main advantage is not accuracy, but speed. This property is what matters for any future real-time application, and in this application the trained LANN dispatch is about 500 times faster than re-solving the QP.

The forecast-error robustness result (section 5.5.2) shows a very important aspect on dispatch cost. It represents the marginal value of a more accurate forecasting for specific cost minimization tasks relative to known fuel availability. The result should be analyzed through the lens of the two most relevant previous works. However, Janefar et al. [7] show that on Bangladesh thermal-only ELD problems PSO is the best cost-and-loss minimiser in the comparison of ACO and Harmony Search. However, ACO is marginally ahead of PSO in this study, and both clearly ahead of GA. The slight difference could be explained by the larger, mixed-source search space from the vast ten-sourced renewable-import combination which added availability constraints rather than a contradiction between the results. This study's result of a change in generation mix to gas and coal (Section 5.2.2, Table 5.4) is consistent with the installed generation mix projected by Olsson and Gardumi in a planning study of Bangladesh using OSeMOSYS [22] that projected higher emissions with cost-only minimization compared to the installed generation mix as a result of applying an emissions constraint.

Finally, it is important to explicitly mention some of the caveats that limit how much the result should be generalized. This study aggregated the fleet into ten source categories and reduced the possibility of including plant-level min-max start-up costs. Again, all the import cost data in the base case document is all-in contracted cost data and does not represent marginal energy costs, addressed in the energy-only sensitivity scenario. Also, the model uses a quadratic cost coefficient, which is a regularisation tested rather than a parameter measured, and the entire optimisation layer is evaluated with respect to a fixed demand series either from history or a forecast, not a live, adaptive dispatch process. None of these reservations affects the major conclusion, yet each places a limitation upon the extent to which these can be considered as the scope of further improvements.

Chapter 6

Conclusion

6.1 Summary of Findings

The aim of this thesis was to develop an integrated solution for the national grid in Bangladesh; first to predict the hourly electricity demand of the grid and second, downstream and without changing the hourly electricity demand prediction model, to assign generation to ten electricity source categories in the grid at minimum cost. Both halves were completed and validated successfully.

- **Forecasting.** A chronological hold-out split and walk-forward evaluation scheme were used to train and compare seven classic machine learning models as well as seven deep learning architectures. Constantly, linear baseline and deep learning architectures were being outperformed by tree based ensemble models. From all the models, Random Forest scored the lowest MAPE (4.16%, $R^2=0.83$) in average walk forward test. LightGBM was almost similar (4.19%). Optuna-based hyperparameter tuning helped to achieve 4.30% on the extended chronological split which improved roughly at 2.7% test MAPE on four year custom training fold. This supports the hypothesis that longer and more recent window better capture the post pandemic demand growth of Bangladesh.
- **Economic load dispatch.** A ten-source, hourly economic dispatch model was constructed from all of the official, publicly available BPDB data, and a certified global optimum (a strictly convex QP with zero KKT stationarity residual, SLSQP cross-checked) was found. Five further approaches including LP, GA, PSO, ACO and a Lagrangian neural network were compared to this certified optimum, on the same problems, with ACO (0.0003% mean gap) and PSO (0.0033%) being the closest, followed by LANN (0.0094%) close behind but roughly 500times faster at inference, and GA was clearly the weakest (0.48%). All the rankings are statistically significant (Friedman $p < 10^{-12}$; Holm corrected p for all pairwise Wilcoxon comparisons $< 10^{-4}$) and they are reproduced on an independent IEEE 30 bus benchmark.
- **Headline cost result.** The reduction in costs of the dispatch re-allocated under the cost assumptions adopted with the help of BPDB's own reported fuel availability limitation is 21.27% (base case) and has a defensible range of 11.39-34.63% depending on the amount of fuel availability assumed. The range is based on BPDB's daily reported fuel availability, not in an unexamined assumption, and is reported

throughout as a model-estimated cost reduction under stated assumptions, not an audited or guaranteed real-world cost reduction.

- **Robustness.** The results were not sensitive to any of the other modelling assumptions that were tested. The quadratic cost coefficient when tested against two independent empirical anchors and varied over a five-point grid, the change in the reduction was only a few percentage points. The minimum-generation rule (-0.27 percentage points) and the forecast error, itself (never more than 0.75% of total cost), over a broad range of forecast errors shows a genuine finding of forecast accuracy for this particular cost-minimisation task at this system’s operating point.

6.2 Contributions to the Field

- An end-to-end framework, directly linking a validated demand forecast and a certified economic dispatch solution for Bangladesh’s national grid, reproducible, single notebook and on the same national dataset - to the authors knowledge, previous studies in Bangladesh have only worked on demand forecasting or economic dispatch separately.
- A new public-data-based fuel-availability constraint for model construction of Bangladesh ELD has been created, based on parsing 178 of BPDB’s own daily generation reports. This gave an otherwise indefensible estimate of cost-reduction (34.6%) a bound on both sides and a base case (21.3%), in the form of an interval (11.4-34.6%). It is an extension of the daily-report scraping technique of Hossain et al. [48] for an earlier period of January-December 2012 with additional fuel-cost fields not previously published.
- An existing comparison of classical six methods, metaheuristic and learned dispatch solvers - LP, a KKT-certified QP, GA, PSO, ACO and an augmented-Lagrangian neural network - on both a real, Bangladesh-specific problem and the IEEE 30-bus benchmark. The additional learned methods and larger renewables extend the existing Bangladesh focused PSO, ACO and Harmony Search comparison study by Janefar et al [7].
- Conditioning on the availability bounds $[lo, hi]$ in addition to the demand - as shown by the LANN’s gap decreasing from 9.4% to 0.0094% when conditioned on $[lo, hi]$ and demand - is a demonstrated, quantified instance of a general lesson for learning-to-optimize methods, with the amortised optimiser being conditioned on the full problem instance.
- A set of parameters from the project designated as having a known source (all capacities, costs and availability parameters marked as officially reported, derived or assumed), plus an auditorial results bundle (all headline results traceable to an exported table), as examples of how this type of downstream, data-driven power-systems study can be made independently verifiable.

6.3 Limitations

Ten-category aggregation. The dispatch model is based on a ten category fleet representation of the fleet, and so cannot be used to model unit commitment, start-up/shut-down costs, per plant heat-rate curves etc. A subset of BPDB’s own gas fleet (approx 4278MW) is used as a substitute for approx 11000MW of installed gas capacity, to determine the cost coefficient for the gas.

Assumed convexity (not measured). The quadratic cost coefficient (κ) is not a directly measured Bangladeshi parameter, but rather is a deliberate regularisation which is required for the solution to be optimal (i.e. to reach a minimum for the total cost), for the marginal price and for the solution to be non-degenerate. Hence allow comparison among methods. It is subject to two separate anchor tests and swept over a range of sensitivities but is still an assumption.

All-in import costs (base case). As to the cost basis, the base case basis is the contracted all-in price for cross-border imports that includes capacity payments which are unavoidable by the dispatch model. An energy-only cost basis is reported as a sensitivity but not as the headline, it still an assumption.

No unit commitment or ramp-coupling across hours. Each hour is dispatched independently; there were no violations of the ramp limits, but they were checked ex post. There were no constraints on minimum generation limits, although they were tested as a sensitivity (a -0.27 percentage point impact). No constraints were placed on minimum generation limits as in each hour they were tested and found to have no impact if they were broken. No constraints were placed on ramp limits, they were checked ex post, but never broke the per-hour decomposition that all methods in this study relied on.

No emissions or environmental constraints. The objective of the dispatch is only to minimise costs; note that the reallocation makes gas and coal more of the energy mix, and the study does not make any claim about the environmental desirability of the dispatch optimised.

Retrospective (not live) evaluation. All optimisation results are post optimisation in terms of historical and/or forecast demand. It has not been tested against an actual, adaptive dispatch process or any feedback loop.

Single-country, single-window scope. All optimisation results are for the dispatch in January-June 2025 and based on cost data from 2024-25 only and cannot be generalized for other periods, seasons (excluding the second half of the year), or any other grid outside Bangladesh.

6.4 Recommendations for Future Work

While this study’s aim to provide a ”cost-efficient resource allocation framework” is a different future work item from those it aims to take beyond into the future, the following are just extensions of the study’s own future work items and the real future-work items that are already identified at the forecasting stage

Make hyperparameter optimisation applicable to the walk-forward setting. For each of the walk-forward folds separately, apply the tuning that, using the relation described above, is obtained through the Optuna framework. To set the two experiments on the same footing, tune each one separately to each walk-forward fold.

Richer forecasting features. Continue to add weather parameters i.e., temperature, humidity, solar irradiance, and try out some other auto-regressive terms that could be useful,

like year on year growth of demand, and an 8,760 hour (yearly) lag.

Deeper forecasting architectures. Consider and systematically test transformer-based time-series models (e.g. PatchTST, Informer) versus the gap with a tree-based ensemble shrines would diminish with a more appropriate architecture search.

Ensembling. Use LightGBM, Random Forest and LSTM predictions and fit the weights of the meta-learner separately using the MAPE for the validation set, instead of its measure calculated on the test set as in previous human biased/errored attempts.

Emissions-aware dispatch. Now, add a carbon price or an explicit emissions cap to the economic dispatch objective, including exactly half of the above with six method comparison analysis, to measure the cost/emissions trade-off which this study would not be able to solve with the cost-only formulation.

Plant-level disaggregation. Expand the 10 category model into rows for individual generating units, to get a cost-of-fuel and cost-of-other-operation-and-maintenance available for the individual plants and a relatively large number of units with that split (around 6,100MW with an available split).

Two zone / network-aware dispatch. BPDB is already in use, and is currently reporting the daily flows of East-West interconnector; the additional network extension easiest to implement is a two-zone model, as per the existing single-bus formulation.

Optimiser LRS implementation at a real time manner. On the LANN with the inference time advantage of 500x versus re-solving the QP, consider the possibility that it can serve as a fast approximate dispatch occurrence tool; periodic as needed to check accuracy; in a near real-time or rolling horizon context.

External validation. Reapply the same framework to another study window (e.g. the second half of 2025), to another study window and/or return to the same framework applied to a different country's grid (e.g. different country but the same study window), to test to what extent the result is Bangladesh- and seasonspecific.

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